Digital Twin-Driven Deep Reinforcement Learning for Energy-Aware CNC Milling Parameter Optimization

ME6324 – AI in Manufacturing

Course Project Report

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This report is generated from the project repository at `G:/My Drive/IITM/AI in Mfg`. All results are reproducible from the committed artifacts using the provided scripts.

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1 Introduction

1.1 Problem Statement

CNC face milling is a fundamental manufacturing operation where a rotating cutter removes material from a workpiece in multiple passes. The cutting parameters – spindle speed (n), feed per revolution (f_r), and axial depth of cut (a_p) – must be chosen to balance four competing objectives:

1. **Productivity:** Maximize material removal rate (MRR $= v_c \cdot a_p \cdot f_r$)
2. **Energy Efficiency:** Minimize total electrical energy consumption (E_{total}), accounting for both machine idle power and cutting power.
3. **Surface Quality Assurance:** Strictly adhere to finished workpiece roughness requirements ($R_a \leq 1.6$ μm).
4. **Tool Wear and Breakage Avoidance:** Limit cumulative flank wear ($V_B \leq 0.3$ mm) and mitigate catastrophic tool failure risks.

Traditional approaches use fixed parameters from handbooks or static optimization (e.g., NSGA-II) that pick one parameter set for the entire batch. These cannot adapt to the *state* of the process – tool wear accumulates, workpiece geometry changes, and material properties vary between batches.

1.2 Proposed Approach

This project develops a **Digital Twin** of the milling process – a high-fidelity physics-based simulator grounded in empirical manufacturing literature – and trains a **Deep Reinforcement Learning (DRL)** agent to dynamically optimize cutting parameters pass-by-pass. The agent observes the current state (flank wear, remaining stock, previous parameters) together with the context it is optimizing for (the energy weight and the material), and selects the next pass parameters to maximize a reward that balances throughput and energy while respecting constraints.

The headline result is a boundary with a number on it. Inside a perfect digital twin the classical receding-horizon controller and the learned policy are comparable, and the controller is the better operator as long as the setup's model of the machine is accurate. Two sweeps of the regime map put a number on where that stops being true: once the flank wear measurement carries a Gaussian error of about **2% of its range**, or once the setup's belief about the material is miscalibrated by more than about **2%**, the learned policy becomes the better operator. The two sweeps are one statement, and the result is actionable: it says how wrong the setup's own model of the machine has to be before the learned policy is worth deploying (Section 7.7).

Key innovations:

- **Speed-dependent specific cutting energy** $k_c(v_c) = k_{c0}(v_c/v_{\text{ref}})^m$ creates a genuine trade-off: faster cutting increases energy per unit volume but reduces idle energy.
- **Domain randomization** during training: material/tool coefficients, workpiece geometry, and initial wear are sampled per episode from literature-calibrated ranges.
- **One environment, one policy:** The energy weight w_2 and the material context live in the observation, so a single PPO policy covers the whole weight range and a whole distribution of materials.

- **A fair classical baseline:** A part-level controller that maximizes the expected episode reward over the same physics, with the breakage risk, the step penalty and the completion bonus priced.
- **Robustness experiment:** The same two arms on the same scenarios in three regimes: the controller with every advantage, the controller with a wrong model of the material, and the controller with a noisy wear measurement.
- **Regime map:** Two sweeps of the same experiment, the measurement noise on the flank wear and the miscalibration of the material model, put a number on the boundary between the two operators: whenever the setup's model of the machine is wrong by more than about **2%**, the learned policy is the better operator, and below it the classical controller is (Section 7.7).
- **Reward sensitivity experiment:** The reward pays the removal rate of every pass, so the credit a plan collects grows with the number of passes it takes, and the measurement shows that part of the gap in the ideal case is an artifact of that accounting (Section 4.5).
- **Batch experiment:** A second environment in which one tool cuts a batch of parts, the tool life is shared across the parts and the tool change is priced.
- **Transfer experiment:** Measures how much a policy trained on one material knows about a harder, faster-wearing material.

1.3 Contributions

1. A Gymnasium-compliant milling environment with explicit physics, energy-aware reward, and domain randomization.
2. A batch environment in which one tool cuts a batch of parts, the tool life is shared across the parts and the tool change is priced.
3. A trained PPO policy evaluated via 100-scenario Monte Carlo sweep with 95% confidence intervals.
4. Energy-weight ablation study (7 weights $\times$ 3 seeds) mapping the empirical Pareto frontier.
5. Comparison against 5 classical baselines, all on the same episodes:
   Constant-High, Constant-Safe, Rule-Based, NSGA-II Static, Receding-Horizon.
6. A robustness experiment against the receding-horizon controller in three regimes: the controller with every advantage, the controller with a wrong model of the material, and the controller with a noisy wear measurement.
7. A regime map with two sweeps, the measurement noise on the flank wear and the miscalibration of the material model. Together they are one statement: the learned policy is the better operator whenever the setup's model of the machine is wrong by more than about **2%**, whether that is a wear measurement noisier than 2% of its range or a material model miscalibrated by more than 2%. The controller's tolerance of both is 2% to 3%, and the two degrade differently: over the noise sweep the controller loses 31% of its reward and the policy 5%. The policy has a limit of its own, around 20% to 25% of model error, and that limit is set by the randomization band of its training.
8. A batch experiment: the batch context is worth 24% fewer tool changes and 9.5% less energy at the same completion, and the per-part controller needs 207.5 s of machine time for a batch of ten parts against the batch-aware policy's 113.1 s.

9. A reward sensitivity experiment: the reward pays the removal rate of every pass, so the credit a plan collects grows with the number of passes it takes, and the measurement shows that the gap in the ideal case is partly an artifact of that accounting (Section 4.5).

10. Transfer experiment on a harder alloy: a policy trained across a distribution of materials is the best arm there without any adaptation.

11. A controller initialization experiment whose result reversed: the initialization is now harmful rather than harmless (Section 7.9).

12. Interactive Streamlit workstation with 2D stock depletion, micro-topography, G-code generation, and multi-controller comparison.

13. Full reproducibility: all results regenerated from committed artifacts by `experiments/summarize_results.py`.

2 Related Work

This project positions itself against the papers held in `docs/papers/`:

Table 1: Position against the reference papers held in `docs/papers/`

Capability	Mo et al. (2025) *J. Manuf. Proc. 156*	Wang et al. (2025a) *J. Manuf. Sys. 78*	**This Work**
Dynamic parameter optimization	✓	✓	✓
Tool wear in the state	✓	✗	✓
Surface roughness constraint	✓	✗	✓
Energy as an objective	✗	✗	✓
Speed-dependent cutting energy	✗	✗	✓
Domain randomization	✗	✗	✓
Conditioned policy (single → frontier)	✗	✗	✓
Transfer experiment across materials	✗	✗	✓
Receding-horizon baseline	✗	✗	✓
Robustness against a model-based controller	✗	✗	✓
Crossover against that controller	✗	✗	✓
Monte Carlo robustness (100 scenarios)	✗	✗	✓
Interactive digital twin workstation	✗	✗	✓

Both reference papers treat energy as related work rather than as an optimization target: in each of them energy appears in the discussion of prior art, not among the objectives. Mo et al. optimize the surface roughness and the production time of a multi-pass turning operation and carry the tool condition and the tool damage in the decision variables, so the wear and the roughness marks carry over. Wang et al. (2025a) build their state from sequences of machining parameters and cutting force features, and their objectives from the dimensional error of the machining features and the machining time; neither the tool wear nor the roughness enters their formulation, so only the dynamic parameter optimization carries over. The third paper in `docs/papers/` (Wang et al., 2025b) solves the discrete step sequencing problem, so continuous cutting parameters are outside its scope. None of the three carries a Taylor wear model, so the wear coefficient of this work is anchored on the tool life that Wang et al. report rather than on a wear equation taken from a paper (Section 3.2).

The speed-dependent cutting energy model (Section 3) is the key enabler here: without it the energy of a part is fixed by the volume removed, both objectives point the same way, and no weight on the energy term can open a trade-off. Putting the energy weight and the material

context in the observation reduces the ablation cost from 21 training runs to one. The receding-horizon controller provides a fair classical benchmark that uses the same physics, re-optimizes at every pass and maximizes the same expected episode reward, and the robustness experiment of Section 7.6 puts the two on the same scenarios in the three regimes a plant actually runs in. The regime map of Section 7.7 then measures where the boundary between the two operators lies, in two sweeps of the setup's own model of the machine.

3 Digital Twin Physics

All equations are implemented in `MillingDT/config/physics.py` and used by the environment, baselines, simulator, and evaluation scripts – a single source of truth. The state the agent sees is the 11-dimensional observation of Section 4.2: eight physics features followed by the conditioning context, and the context is present in every task, so the equations below are the same equations for every task.

3.1 Geometry and Kinematics

Cutter diameter $D = 50$ mm, cutting length $L_{\text{cut}} = 100$ mm, total machining allowance $\kappa_{\text{total}} = 5.0$ mm.

$$v_c = \frac{\pi D n}{1000} \qquad [\text{m/min}] \tag{1}$$

$$\text{MRR} = v_c \cdot a_p \cdot f_r \qquad [\text{mm}^3/\text{min}] \tag{2}$$

$$t_{\text{cut}} = \frac{L_{\text{cut}}}{n f_r} \qquad [\text{min}] \tag{3}$$

$$N_{\text{passes}} = \left\lceil \frac{\kappa_{\text{total}}}{a_p} \right\rceil \qquad [\text{integer}] \tag{4}$$

3.2 Tool Wear (Extended Taylor Model)

Flank wear rate per pass:

$$\Delta V_B = C_{\text{wear}} \cdot v_c^{\alpha} \cdot f_r^{\beta} \cdot a_p^{\gamma_{\text{wear}}} \cdot t_{\text{cut}} \tag{5}$$

with calibrated coefficients $C_{\text{wear}} = 1.0 \times 10^{-2}$, $\alpha = 0.3$, $\beta = 0.2$, $\gamma_{\text{wear}} = 0.15$. Cumulative wear $V_{B,k} = \sum_{i=1}^{k} \Delta V_{B,i}$.

The coefficient is anchored on a tool life, not on a wear equation. None of the three reference papers carries a Taylor wear model, so there is no published wear equation to calibrate the coefficient against. What Wang et al. (2025a) do report is a tool life: 40 parts cut with 5 identical tools, which is about 8 parts per tool. The coefficient is set so that the tool life at the reference cutting condition is about 9 parts per tool, which puts the tool life inside the range where a batch can run out of a tool. The previous value $C_{\text{wear}} = 1.1 \times 10^{-4}$ gave a tool life of about 870 parts, so the wear constraint did nothing inside a part and the tool life was irrelevant to a batch; the wear model was inert.

The exponents are the exponents of the wear rate, not of the Taylor tool life. The classical tool life equation carries the exponents $(4.0, 0.8, 3.2)$, and with them the tool life spans six orders of magnitude over the machine envelope, which is not a usable envelope either. The exponents used here describe how fast the wear accumulates, so the tool life varies only by a factor of a few across the envelope: a faster cut barely shortens the tool life in this model. That is a known weakness of the wear model and it is carried into the limitations of Section 10.2. It is the reason the wear constraint acts mainly on the aggregate of a batch rather than on the choice of a single pass.

3.3 Surface Roughness (Empirical Regression)

$$R_a = k_1 f_r^2 + k_2 V_B + \frac{k_3}{v_c} \tag{6}$$

with $k_1 = 12.5$, $k_2 = 0.8$, $k_3 = 0.5$. The finished surface (last pass) must satisfy $R_a \leq R_{a,\max} = 1.6\ \mu\text{m}$.

3.4 Energy Consumption

Specific cutting energy grows with cutting speed:

$$k_c(v_c) = k_{c0}\left(\frac{v_c}{v_{\text{ref}}}\right)^m, \quad P_{\text{cut}} = k_c(v_c) \cdot \text{MRR}, \quad P_{\text{total}} = P_{\text{idle}} + P_{\text{cut}} \tag{7}$$

with $P_{\text{idle}} = 500$ W, $k_{c0} = 2.5\ \text{W} \cdot \text{min/mm}^3$, $m = 0.6$, v_{ref} from the reference cutting condition. Energy per pass: $E_{\text{pass}} = P_{\text{total}} \cdot t_{\text{cut}}$. Total energy per part: $E_{\text{part}} = \sum_{k=1}^{N_{\text{passes}}} E_{\text{pass},k}$.

Proposition 1 (Interior Minimum)**.** *With $m > 0$, the energy per part $E_{part}(v_c)$ has an interior minimum in v_c. The cutting term rises with v_c while the idle term falls, creating a genuine trade-off that the energy weight w_2 can select.*

This is validated by test `test_part_energy_has_an_interior_minimum_in_speed`.

3.5 Stochastic Tool Breakage

Probability of catastrophic breakage per pass, applied only once the flank wear passes a threshold:

$$p_{\text{break}} = 0.3 \cdot \underbrace{\frac{V_B - 0.15}{V_{B,\max} - 0.15}}_{\text{wear factor}} \cdot \underbrace{\left(\frac{\text{MRR}}{\text{MRR}_{\max}}\right)^2}_{\text{load factor}} \qquad (V_B > 0.15\ \text{mm},\ \text{else } 0) \tag{8}$$

where $\text{MRR}_{\max}$ is the removal rate of the machine envelope at its upper bounds. The draw comes from the environment's own generator, which `reset(seed=...)` seeds, so a scenario replayed with the same seed breaks in the same place. If triggered, the episode terminates with a -100 reward.

4 Environment Design

4.1 Registrations

Two Gymnasium environments are registered in `MillingDT/env/__init__.py`. The per-part one is exported as the constant `MillingDT.env.ENV_ID` and the batch one as `MillingDT.env.BATCH_ENV_ID`:

1. `MillingDTEnv-v1`: the part environment. The observation is always $\mathbb{R}^{11}$: the eight physics features followed by the conditioning context (the energy weight w_2, the wear coefficient C_{wear} and the roughness coefficient k_1). The flags `randomize` and `randomize_weight` decide whether the scenario and the energy weight are drawn per `reset()`. A task with a pinned weight is the same environment with a constant in the context, so there is a single observation shape across the whole project.

2. `MillingDTBatchEnv-v1`: the batch environment. One episode is a batch of ten parts cut with one tool. The observation is $\mathbb{R}^{13}$: the eleven features of the part the tool is cutting, then the fraction of the tool life that is left and the fraction of the batch that is still to cut. The tool is replaced when it breaks or reaches its wear limit, the wear of the new tool

starts at zero, and the replacement costs 500 reward units against the 200 a finished part is worth. The physics is not duplicated: the batch environment wraps the part environment and adds the tool accounting.

Two optional arguments make the environment behave like a plant rather than like a simulator with perfect knowledge. The argument `belief` sets what the setup *believes* the material is, while the physics keep running on the truth, and the argument `obs_noise` puts noise on the two measured state features, the flank wear and the remaining stock. Both are used by the robustness experiment of Section 7.6.

4.2 State Space

Table 2: Observation (11 dimensions), every feature mapped onto $[0, 1]$

Index	Feature	Normalization
0	Flank wear V_B	tanh squash over $[0, V_{B,\max}]$
1	Remaining stock κ_{rem}	tanh squash over $[0, \kappa_{\text{total}}]$
2	Previous spindle speed n	$(n - n_{\min})/(n_{\max} - n_{\min})$
3	Previous feed f_r	$(f_r - f_{\min})/(f_{\max} - f_{\min})$
4	Previous depth a_p	$(a_p - a_{p,\min})/(a_{p,\max} - a_{p,\min})$
5	Previous MRR	$\text{MRR}/\text{MRR}_{\max}$, clipped
6	Previous roughness R_a	tanh squash over $[0, R_{a,\max}]$
7	Tool broken	1 if broken, else 0
8	Energy weight w_2	$w_2 \in [0, 1]$
9	Wear coefficient C_{wear}	over the scenario band $[0.6, 1.4]$ of nominal
10	Roughness coefficient k_1	over the scenario band $[0.8, 1.2]$ of nominal

Every feature is normalized against the *scenario's own* limits, so the observation is scale free across geometries and materials. The tanh squashing keeps the gradient magnitude when a constraint is violated, instead of clipping the value to 1.

The context occupies the last three slots in every task. A policy trained with a pinned weight sees a constant there, a policy trained with the weight drawn sees the weight it is being asked for, and a policy trained across the material distribution sees the material it is cutting. The shape never changes, so one trained policy can be asked for any weight and any material at runtime. The two material slots carry what the setup *believes* the material is, not what it is: a plant that does not know its material passes a belief that differs from the truth while the physics keep running on the truth. Of the eight state features only two are measured, the flank wear and the remaining stock, and only those two carry the measurement noise.

4.3 Action Space

Continuous 3D box $\mathcal{A} = [-1, 1]^3$, mapped to physical parameters:

$$n = n_{\min} + \frac{a_1 + 1}{2}(n_{\max} - n_{\min}) \tag{9}$$

$$f_r = f_{\min} + \frac{a_2 + 1}{2}(f_{\max} - f_{\min}) \tag{10}$$

$$a_p = a_{p,\min} + \frac{a_3 + 1}{2}(a_{p,\max} - a_{p,\min}) \tag{11}$$

The policy output is unbounded; the environment clips to $[-1, 1]$ before mapping, guaranteeing the machine envelope is never violated.

4.4 Reward Function

The per-step reward is:

$$
\begin{aligned}
r = \; & \underbrace{100\left(w_1 \frac{\mathrm{MRR}}{\mathrm{MRR}_{\max}} - w_2 \frac{E_{\mathrm{pass}}}{E_{\mathrm{ref}}}\right)}_{\text{Objective}} + \underbrace{20\,\Delta\left(\tfrac{\kappa_{\mathrm{removed}}}{\kappa_{\mathrm{total}}}\right)}_{\text{Progress}} \\
& - \underbrace{50\,\mathbb{1}[V_B > V_{B,\mathrm{safe}}]}_{\text{Wear penalty}} - \underbrace{100\,\mathbb{1}[\text{broken}]}_{\text{Breakage}} \\
& - \underbrace{50\,\mathbb{1}[\text{finishing pass} \wedge R_a > R_{a,\mathrm{safe}}]}_{\text{Roughness penalty}} - 0.05
\end{aligned}
\tag{12}
$$

plus, on a part finished inside spec, a +200 completion bonus and a Pareto margin bonus of at most +5:

$$
\mathrm{HV} = \max\!\left(0, 1 - \tfrac{T_{\mathrm{total}}}{T_{\mathrm{ref}}}\right) \cdot \max\!\left(0, 1 - \tfrac{R_a}{R_{a,\mathrm{safe}}}\right) \cdot 5 \tag{13}
$$

All of these magnitudes live in the `reward` section of `MillingDT/config/physics_constants.yaml`, so they are in one place and the file states that they are a design decision. **The scale of the objective term matters as much as the weights.** The objective term is what decides the operating point, so it has to be worth a sensible fraction of the value of a finished part. With the objective scale at 5 the whole throughput and energy trade-off of a part was worth about 1.2 points of reward against a completion bonus of 200, so a policy that finished every part at any cost scored well and the energy weight had almost nothing to select. At 100 the trade-off is worth about fifty points per part, a quarter of the value of a part, and the completion bonus still dominates the decision to finish. The constraint penalties (50 for a pass above the wear safety limit, 100 for a broken tool, 50 for a finishing pass outside the roughness safety limit) sit above the progress bonus of 20, so that breaking a constraint is never worth a throughput or energy gain.

$R_{a,\mathrm{safe}} = 0.85\,R_{a,\max}$, $w_1 + w_2 = 1$, and E_{ref} is the energy of a pass under the reference condition, recomputed for every scenario from that scenario's diameter, cutting length and energy coefficients. The reward is validated by `test_reward_matches_an_independent_calculation`, which recomputes it from `MillingDT/config/physics.py`, independently of the code that assembles it, and asserts agreement to 10^{-9}.

4.5 The Reward and the Accounting of a Rate

The reward pays the objective term once per pass, and the objective term carries the removal rate MRR of that pass. That is the standard way to express a rate in a reward that is paid per pass, and it has a consequence that has to be measured rather than argued about: the total credit a part collects grows with the number of passes the plan takes. A plan that cuts the same part in more, shallower passes is paid more for it.

The measurement was made on the same 100 randomized scenarios at $w_2 = 0.4$ that carry the robustness experiment, by `experiments/reward_sensitivity.py`. The script rolls both arms out and accumulates the reward terms in a form that lets the accounting be varied afterwards, so the same rollouts are scored under every objective scale and under every objective form.

Table 3: What the two arms collect on the same 100 scenarios at $w_2 = 0.4$. The objective credit is the sum of the per-pass removal rates that the reward pays, and the mean reward is the episode reward of the arm.

Quantity	Controller	Policy
Passes per part	5.85	4.00
Objective credit collected	+0.413	+0.326
Mean reward	**248.9**	233.6

The two plans remove the same stock allowance. The controller takes more passes, so it collects **27% more objective credit** for the same material, and at the objective scale of the configuration that is about nine of the fifteen reward points that separate the two arms in the ideal case. The ideal case comparison is therefore biased against the learned policy, and the bias belongs to the reward, not to either method.

Three things were checked before this was written up as a limitation of the comparison.

1. The objective scale is not the lever. The controller wins at every scale from 5 to 4000, and the gap only grows with it: 7.1 points at 5, 10.9 at 50, 15.3 at 100, 23.9 at 200, 41.2 at 400, 93.0 at 1000 and 352.0 at 4000. The sign never flips. The table is Table 11 and the artifact is `results/evaluation/reward_sensitivity.csv`.

2. The part level alternative was implemented and rejected. Crediting the material of the part against the energy of the part removes the pass count from the accounting, and under that form the two arms are much closer: a 5.0 point gap at the scale of 100, against 15.3 for the shipped form, and the policy is ahead at the large scales, by 9.4 points at 1000 and 57.4 at 4000. It is not the right fix, because the material a part gives up is fixed by its stock. The volume of a part is $\pi D \kappa_{\text{total}} L_{\text{cut}}/1000$, independent of every cutting parameter, so the throughput weight would then be inert: the reward would be an energy objective with a throughput weight that does nothing, and the weight sweep of this project would stop measuring what it claims to measure.

3. What a correct formulation would need. The removal rate of a part is the ratio of a fixed volume to the part's time, and a rate cannot be accumulated over a part without becoming a volume. A reward that expresses it has to either price the cycle time directly, or compare the part's average rate to a reference at the end of the part. Both change the meaning of the weights, and the second makes the objective sparse and the credit assignment poor. Neither was adopted, and the reward as shipped carries the artifact instead, documented rather than hidden.

The consequence for the reading of the ideal case follows from this. The defensible reading of Table 10 is that the two methods are comparable inside a perfect digital twin: the learned policy is ahead on the physical metrics (22% more material removed, a third less cycle time, 3.7% less energy) and behind on the tool breakage (6% against 3%), and the measured gap on the reward is partly the artifact of how the reward pays for a rate.

4.6 Episode Logic

- `terminated=True`: Part finished ($\kappa_{\text{rem}} \leq 0$), tool broken, or wear exceeded.
- `truncated=True`: Maximum steps (200) reached.
- The roughness penalty is charged once, at the finishing pass, and only when that pass ends outside $R_{a,\text{safe}}$. Intermediate passes carry no roughness penalty; the wear penalty is charged on every pass whose cumulative flank wear exceeds $V_{B,\text{safe}}$.

4.7 Domain Randomization

When `randomize=True`, each `reset()` samples a new scenario from `MillingDT/config/scenarios.py`:

- Workpiece: $\kappa_{\text{total}} \in [2, 15]$ mm, $D \in [40, 150]$ mm, $L_{\text{cut}} \in [50, 200]$ mm
- Material/Tool: C_{wear} at $[0.6, 1.4]\times$ nominal, k_1, k_2, k_3 each at $[0.8, 1.2]\times$ nominal
- Initial wear: $V_{B,0} \sim \mathcal{U}[0, 0.8]\, V_{B,\max}$

The same tables are used for Monte Carlo evaluation, so the policy is measured on the distribution it was trained on. Measured on the trained policy, the eight physics features the policy receives on the two sets agree to 0.041 in the mean of their normalized values, against a standard deviation of 1.

5 Reinforcement Learning

5.1 Algorithm

Proximal Policy Optimization (PPO) with MLP policy, implemented via Stable-Baselines3.

Table 4: PPO hyperparameters

Parameter	**Value**
Policy	MlpPolicy (2 hidden layers, 256 units each)
Learning rate	3×10^{-4} (linear schedule to 0)
n_steps	2048
Batch size	64
n_epochs	10
γ	0.99
GAE λ	0.95
Clip range	0.2
Entropy coefficient	0.02
Value function coefficient	0.5
Max grad norm	0.5
VecNormalize	`norm_obs=True, norm_reward=True`
Environments	4 parallel (`SubprocVecEnv`)
Device	CPU (2.5× faster than GPU for this MLP)

Training uses a custom `ProgressQueueCallback` that sends a heartbeat every 1000 steps to the supervisor (`run_experiments.py`); a dropped heartbeat triggers worker termination.

5.2 Training Runs

1. **Production policy:** 1,000,000 steps, seed 42, `--randomize --randomize-weight`. Both the machining scenario and the energy weight are drawn per episode, so one network covers the whole weight range and the whole material distribution. Model at `results/run_PPO_seed42/`. This is the policy the dashboard, the Monte Carlo sweep, the runtime weight sweep and the baseline comparison all use.

2. **Ablation grid:** 7 weights $w_2 \in \{0.0, 0.2, 0.4, 0.5, 0.6, 0.8, 1.0\}\times$ 3 seeds $\{42, 43, 44\}$, 150,000 steps each, trained in parallel via `experiments/run_experiments.py`. Every ablation policy is trained with its energy weight **pinned** and is evaluated at the weight it

was trained for, so it is always asked the question it was trained for. The model directories follow the pattern `results/run_ablation_w2_0_4_seed42`, one per weight and seed.

3. **Batch-aware policy:** 1,000,000 steps, seed 42, on the batch environment (`--env-id MillingDTBatchEnv-v1 --randomize`). The observation carries the tool life left and the fraction of the batch still to cut. Model at `results/run_PPO_batch_seed42/`.

4. **Policy trained with a noisy wear measurement:** 1,000,000 steps, seed 42, `--randomize --randomize-weight --obs-noise 0.05`. The noise is on the two measured state features, the flank wear and the remaining stock. Model at `results/run_PPO_noisystate_seed42/`.

All training paths use the same script, `experiments/train_rl.py`: `--randomize` draws a new machining scenario per episode, `--randomize-weight` draws a new energy weight per episode, `--w2` pins the weight of the task, `--env-id` selects the environment and `--obs-noise` puts noise on the measured state. The controller initialization experiment trains its own arm through the same script with the flag `--init-from-controller`.

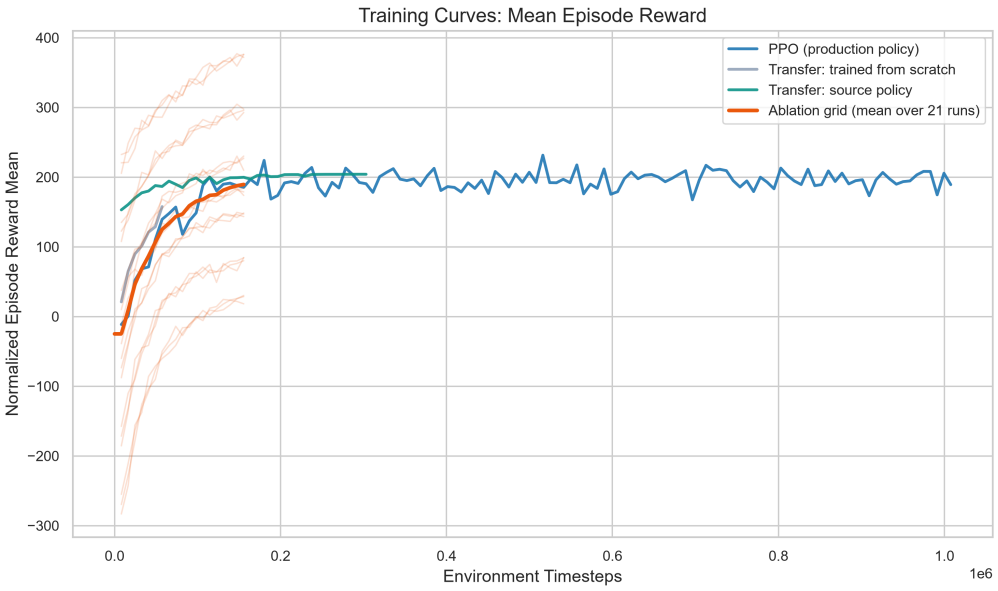


Figure 1: Training curves of every run: the mean episode reward against the environment steps. The production policy in blue, the two arms of the transfer experiment, and the mean over the 21 ablation policies in orange with the individual runs drawn faintly behind it.

Figure 1 shows what the budgets buy. The production policy reaches the level of about 200 inside its first 200,000 steps and then oscillates around it for the remaining 800,000, so the size of the budget buys stability rather than a higher reward. The 21 ablation policies reach the same level within their 150,000 steps, the individual runs drawn faintly behind their mean spread widely, and the two transfer arms converge inside their 300,000 step budget. The vertical axis is the normalized reward mean of `VecNormalize`, so it does not carry the scale of the episode reward reported in the evaluation tables.

5.3 Monte Carlo Evaluation

`python -m experiments.evaluate --scenarios 100` runs the production policy over 100 randomized scenarios (fixed seed 42). Each scenario is a full machining episode. Reports mean ± std and 95% CI (t-distribution) for:

- MRR, Energy/part, Cycle time
- Max R_a, Max V_B
- Constraint violation rates (worst pass vs finished surface)
- Tool breakage rate, Part completion rate

6 Classical Baselines

Five non-RL baselines implemented in `experiments/run_baselines.py`:

1. **Constant-High:** $n = 8000, f_r = 0.35, a_p = 2.5$ (upper bounds).
2. **Constant-Safe:** $n = 2000, f_r = 0.05, a_p = 0.1$ (lower bounds).
3. **Rule-Based Degradation:** Feed at maximum, spindle speed reduced by 20% every second pass, as a shop-floor heuristic for tool degradation.
4. **NSGA-II Static:** Multi-objective optimization (pymoo) picking *one* parameter set for the whole part. Decision variables $[n, f_r, a_p]$. Objectives: maximize MRR, minimize Energy. Constraints: $R_{a,\text{final}} \leq R_{a,\max}$, $V_{B,\text{final}} \leq V_{B,\max}$.
5. **Receding-Horizon Controller:** A vectorized grid search over (n, f_r, a_p) that re-solves at every pass, simulating the remaining passes of the part with the current V_B and κ_{rem}. It is a part-level maximizer of the expected episode reward: for every candidate it prices the objective term, the progress bonus, the wear penalty, the step penalty, the completion bonus, the Pareto margin and the breakage probability, and it keeps the candidate with the best expected value. The plan of a part is two parameter sets, a roughing feed and a finishing feed, because the reward penalizes the roughness of the finishing pass only and a plan that holds one feed for the whole part would have to satisfy the roughness limit on the worst pass. This is the strongest classical baseline.

The controller is a baseline in the strict sense: it optimizes the same quantity the environment pays, over the same physics, and it is given the model and the state it is handed. It is not given anything the policy does not have, and it is not a stand-in for the reward. Its quality is the measure of the comparison: on the same scenarios it went from 184.3 to 248.9 when it was changed from a per-pass maximizer of the objective term into a part-level maximizer of the expected episode reward. All baselines use the *same physics* (`MillingDT/config/physics.py`) and are evaluated on the same 50 episodes of the nominal machining condition, at the same energy weight $w_2 = 0.4$ at which the learned policy is queried.

7 Results

All tables below are regenerated from committed CSVs by
`python -m experiments.summarize_results`.

7.1 Production Policy – Monte Carlo Sweep (100 Scenarios)

Table 5: Production policy Monte Carlo results (100 scenarios)

Metric	**Mean**	**Std**	**95% CI**
Material removal rate (mm^3/min)	1608.49	550.69	[1499.22, 1717.76]
Energy per part (kJ)	66.40	48.03	[56.87, 75.93]
Cycle time per part (s)	11.43	6.93	[10.06, 12.81]
Max surface roughness (μm)	1.610	0.178	[1.575, 1.645]
Max flank wear (mm)	0.1407	0.0672	[0.1274, 0.1540]
Constraint violations			
Roughness violations, worst pass	55.0%		[45.2%, 64.8%]
Roughness violations, finished surface	4.0%		[0.2%, 7.8%]
Roughness > safety limit, worst pass	87.0%		[80.4%, 93.6%]
Roughness > safety limit, finished surface	4.0%		[0.2%, 7.8%]
Flank wear violations ($V_B > 0.3$ mm)	0.0%		[0.0%, 0.0%]
Tool breakage	6.0%		[1.3%, 10.7%]
Parts completed within spec	94.0%		
Scenarios	100		
Mean operating point	7907 RPM, 0.318 mm/r, 2.07 mm, 4.0 passes		

Key observations:

- **The policy is a much more aggressive operator than it was under the old reward.** It removes 1608.5 mm^3/min, against 1205.2 before, and it cuts a part in 11.4 s against 20.3 s. The price is paid in the tool and in the part quality: the breakage rate is 6.0% and 4.0% of the finished surfaces are outside the roughness limit.
- **The roughing passes are rough:** 87% of the worst passes are above the safety margin ($0.85 \times 1.6 = 1.36$ μm) and 55% are above the hard limit of 1.6 μm. The finished surface, which is the quality the reward prices, is inside the limit in 96% of the scenarios.
- **Wear is never the binding constraint on a single part:** 0% wear violations, and the mean maximum wear of 0.1407 mm is under half the 0.3 mm limit. The constraint binds on the aggregate of a batch, which is why the batch experiment of Section 7.8 exists.
- **High variance:** the standard deviation of MRR is 34% of its mean, and the scenarios range from 2 to 15 mm of stock, so the policy is measured on a wide spread of operating points.

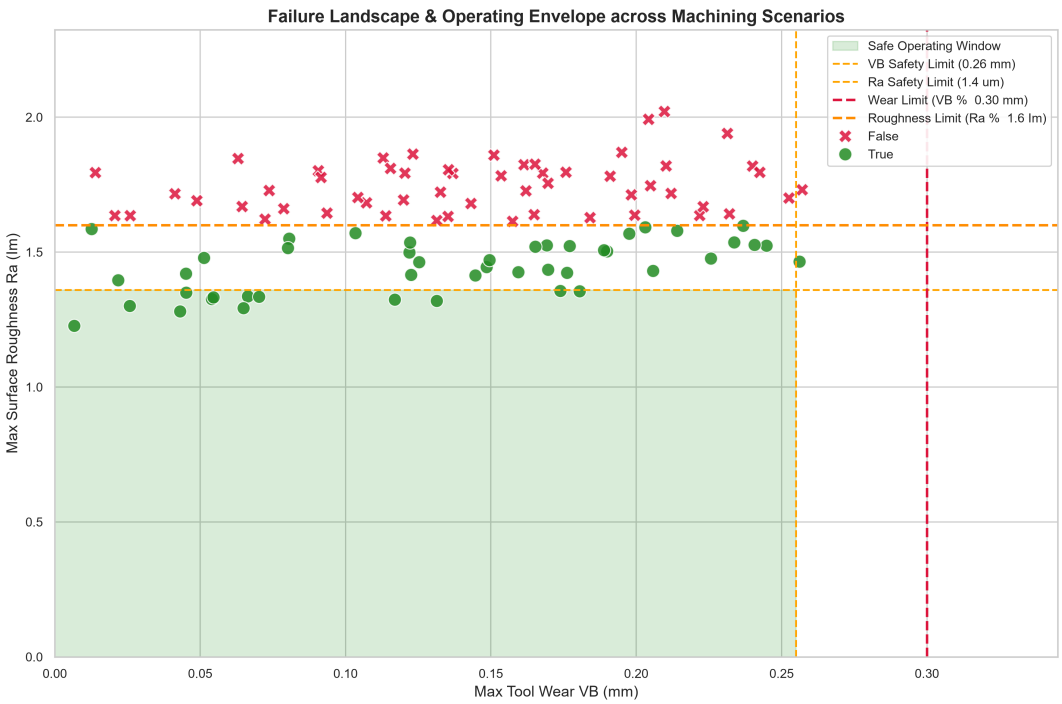


Figure 2: The operating envelope of the 100 Monte Carlo scenarios: the maximum flank wear of a scenario on the horizontal axis against its maximum surface roughness on the vertical axis. The shaded rectangle is the safe operating window, bounded by the wear safety limit $V_{B,\text{safe}}$ (0.255 mm, drawn at 0.26 mm) and the roughness safety limit $R_{a,\text{safe}}$ (1.36 μm, drawn at 1.4 μm). The dashed lines mark the hard limits $V_{B,\max} = 0.3$ mm and $R_{a,\max} = 1.6$ μm. A circle marks a scenario that stays inside both hard limits, a cross a scenario that ends above the roughness limit.

Figure 2 puts the two constraints on one plane and shows where the 100 scenarios land. No scenario reaches the wear limit: the rightmost point stops just inside the wear safety limit at 0.26 mm, well short of $V_{B,\max} = 0.3$ mm, which is the 0% wear violation rate of Table 5. The scenarios lie on a downward slope, the more the tool wears the smoother the surface, so the scenarios that push the roughness up are the ones with the least wear. The 55 crosses above $R_{a,\max} = 1.6$ μm are the worst-pass roughness violations of the table, and the circles are the 45 scenarios whose worst pass stays inside the limit, the ones between $R_{a,\text{safe}}$ and $R_{a,\max}$ exceeding the safety margin without breaking it. The finished surface is a separate quantity, and it is outside the limit in 4 of the 100 scenarios.

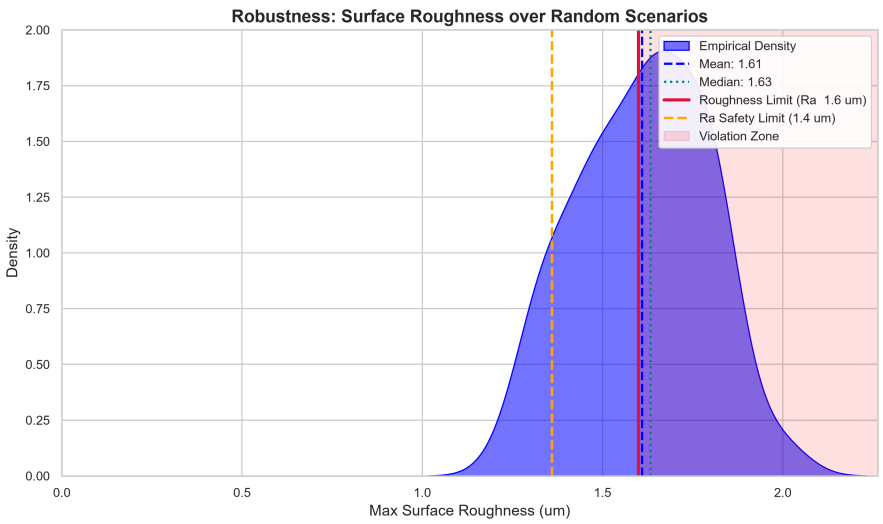


Figure 3: Density of the worst-pass surface roughness over the 100 Monte Carlo scenarios, with the roughness limit $R_{a,\max} = 1.6$ μm, the safety limit $R_{a,\text{safe}} = 1.36$ μm (drawn at 1.4 μm) and the violation zone beyond the limit.

Figure 3 is that distribution in one axis. The mean is 1.61 μm and the median 1.63 μm, and the density is concentrated on and just above the limit, which is the same fact the 55% worst-pass violation rate states. The distribution has a single mode, near the limit: under the old reward it carried a second mode near 0.9 μm from the scenarios cut conservatively, and that mode is gone, because the reward now pays for the throughput that the conservative scenarios give up. The finishing pass is a separate quantity, and it is outside the limit in 4% of the scenarios.

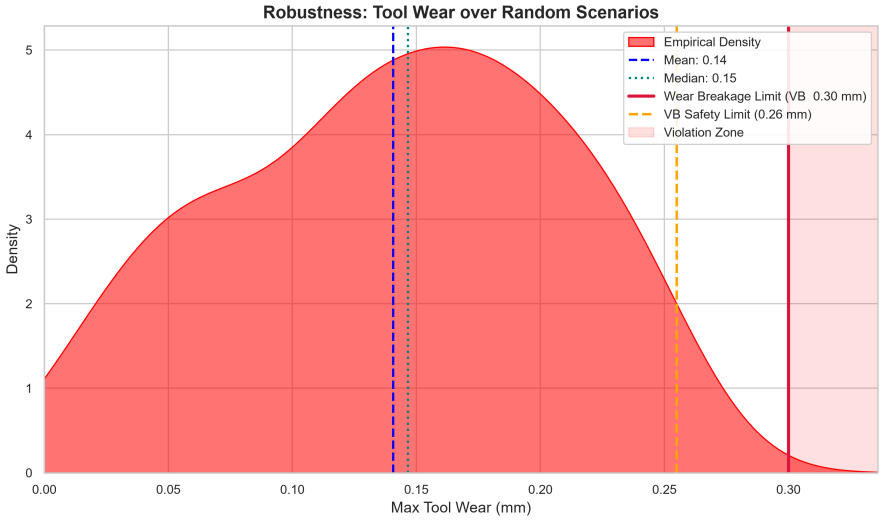


Figure 4: Density of the maximum flank wear over the 100 Monte Carlo scenarios, with the wear limit $V_{B,\max} = 0.3$ mm, the safety limit $V_{B,\text{safe}} = 0.255$ mm (drawn at 0.26 mm) and the violation zone beyond the limit.

Figure 4 is the same view of the wear. The mean sits at 0.14 mm and the median at 0.15 mm, and the density now reaches the safety limit at 0.26 mm instead of collapsing well before it, so the wear of a single part runs close to the limit without crossing it. The 6% breakage rate of Table 5 comes from the stochastic draw above 0.15 mm of wear, not from the wear limit

itself.

7.2 Energy Weight Ablation – Pareto Frontier

Table 6: Ablation study: mean ± std over 3 seeds, 50 scenarios each, every policy trained and evaluated with its weight pinned

w_2	MRR (mm^3/min)	Energy (kJ)	Cycle (s)	n (RPM)	f_r (mm/r)	Frontier
0.0	1397.1 ± 14.8	70.4 ± 0.1	14.8 ± 0.3	7998	0.299	dominated
0.2	1456.9 ± 14.6	70.0 ± 0.1	13.8 ± 0.2	7992	0.299	dominated
0.4	1524.9 ± 40.2	69.3 ± 0.2	13.0 ± 0.3	7896	0.296	dominated
0.5	1556.6 ± 9.7	69.1 ± 0.2	12.7 ± 0.1	7859	0.296	✓
0.6	1519.8 ± 36.4	68.3 ± 0.4	13.1 ± 0.4	7638	0.296	✓
0.8	1224.4 ± 11.3	63.0 ± 0.2	15.4 ± 0.1	6573	0.292	✓
1.0	918.5 ± 14.9	57.2 ± 0.2	20.1 ± 0.4	5382	0.292	✓

Proposition 2 (Frontier Shape)**.** *Across the Pareto frontier ($w_2 \in [0.5, 1.0]$): MRR falls 41% (1557 → 919 mm^3/min) while energy falls 17% (69.1 → 57.2 kJ). The trade-off is asymmetric: the energy of the part is bought with a lot of removal rate, and the spindle speed falls with the energy weight, which is the mechanism the model predicts.*

The dominated points ($w_2 = 0.0, 0.2, 0.4$) all spend more energy than a point on the frontier. The two low weights push the speed to the top of the envelope, which raises $k_c(v_c)$ without cutting the idle energy enough to compensate, and $w_2 = 0.4$ is dominated by $w_2 = 0.5$, which removes more material for less energy.

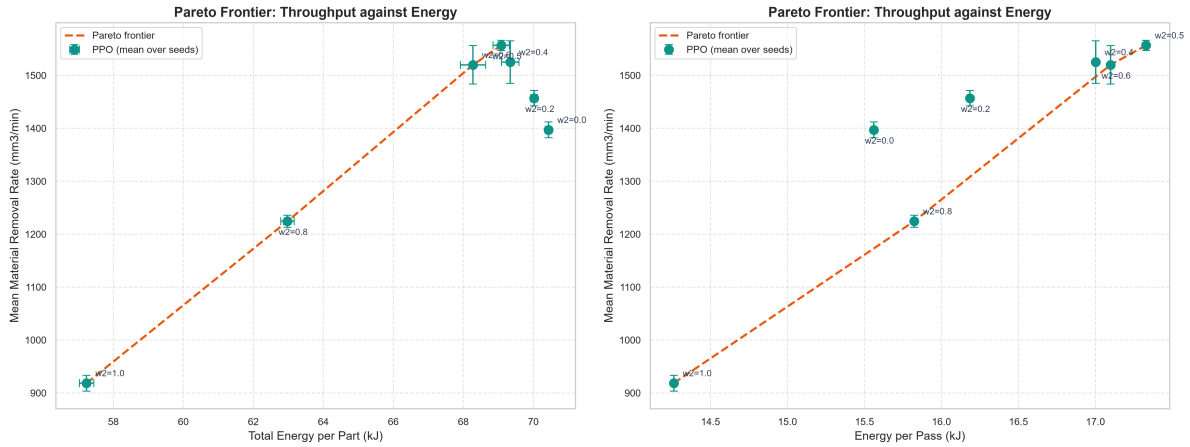


Figure 5: The ablation frontier: the mean removal rate of each of the seven pinned-weight policies against its energy, with the spread over the three seeds as error bars and the non-dominated points joined into the frontier. The left panel carries the energy of a part, the metric of Table 6, and the right panel the energy of a single pass, the quantity the reward charges.

Figure 5 draws the seven policies in the plane of energy against removal rate. The frontier is the chain of four non-dominated points from $w_2 = 0.5$ to $w_2 = 1.0$, and it is the same chain in both panels, so the energy of a pass and the energy of a part select the same operating points here. The three dominated points sit off it: $w_2 = 0.0$ and $w_2 = 0.2$ to the right of the chain, spending more energy for less material, and $w_2 = 0.4$ in the cluster at the top of the chain, just under the segment between $w_2 = 0.5$ and $w_2 = 0.6$.

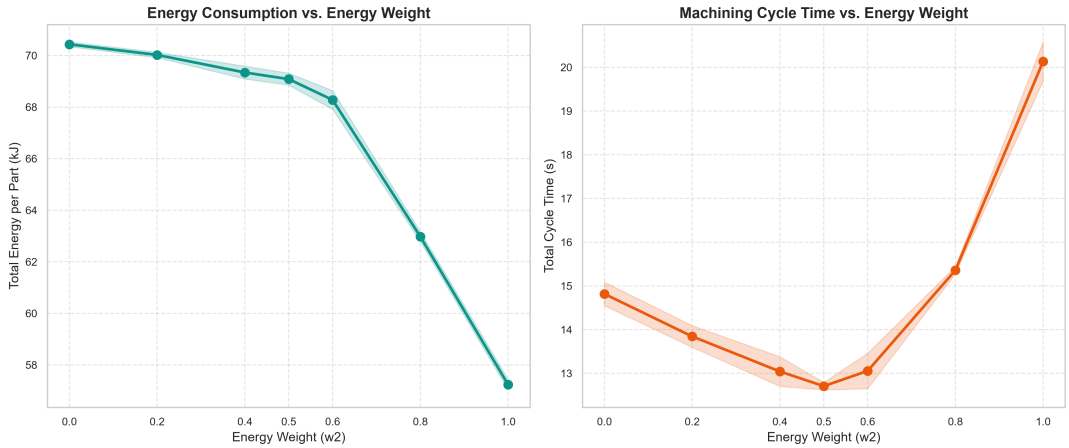


Figure 6: The energy weight sweep of the ablation policies: the energy per part (left) and the cycle time per part (right) against the energy weight, with the spread over the three seeds shaded.

Figure 6 shows the same sweep as two curves against the weight. The energy per part falls from 70.4 kJ at $w_2 = 0.0$ to 57.2 kJ at $w_2 = 1.0$, and the shaded band shows that the three seeds agree on the shape. The cycle time is not monotone: it falls from 14.8 s to its minimum of 12.7 s at $w_2 = 0.5$ and then rises to 20.1 s at $w_2 = 1.0$. Past $w_2 = 0.5$ the energy is bought with the removal rate and with the time, and the two heaviest weights are where the trade-off is steepest. That is the region in which the one conditioned policy loses to the per-weight policies of Section 7.4.

7.3 Runtime Weight Sweep of the Production Policy

The weight is a query, not a retrain: the same policy is asked for each point of the trade-off at runtime (`python -m experiments.evaluate --sweep-weight`) over the same 100 scenarios.

Table 7: Production policy asked for every weight at runtime (100 scenarios)

w_2	MRR (mm^3/min)	Energy (kJ)	Cycle (s)	n (RPM)	f_r (mm/r)	R_a violation rate
0.0	1475.9	68.4	13.1	8000	0.323	56.0%
0.2	1518.0	65.8	11.8	7997	0.323	56.0%
0.4	1608.5	66.4	11.4	7907	0.318	55.0%
0.5	1630.3	66.0	11.4	7798	0.317	54.0%
0.6	1540.8	65.5	12.4	7433	0.317	55.0%
0.8	1114.5	57.9	17.4	5841	0.316	56.0%
1.0	720.4	51.7	26.1	4111	0.315	56.0%

As the weight goes from 0.0 to 1.0 the removal rate falls from 1475.9 to 720.4 mm^3/min, the energy of the part falls from 68.4 to 51.7 kJ, the cycle time rises from 13.1 to 26.1 s and the spindle speed falls from 8000 to 4111 RPM. The operating point moves along the trade-off without a retrain. The response is not monotone in the removal rate: it peaks at 1630.3 mm^3/min at $w_2 = 0.5$, so the weight buys the energy of the first half of the range with throughput and only starts to pay for it after that. The roughness violation rate of the worst pass stays at 54% to 56% over the whole range, so the operating point moves without the constraint behaviour changing.

7.4 One Policy against the Per-Weight Policies

The production policy is compared against the 21 ablation policies, one per weight, each trained and evaluated with its weight pinned (`python -m experiments.evaluate --sweep-weight --compare-only`).

Table 8: One policy against 21 policies, one per weight. Removal rate in mm^3/min, energy per part in kJ.

w_2	MRR, one policy	MRR, per weight	MRR gap	Energy, one policy	Energy, per weight	Energy gap
0.0	1475.9	1397.1	+5.6%	68.4	70.4	−2.9%
0.2	1518.0	1456.9	+4.2%	65.8	70.0	−6.1%
0.4	1608.5	1524.9	+5.5%	66.4	69.3	−4.2%
0.5	1630.3	1556.6	+4.7%	66.0	69.1	−4.4%
0.6	1540.8	1519.8	+1.4%	65.5	68.3	−4.0%
0.8	1114.5	1224.4	−9.0%	57.9	63.0	−8.0%
1.0	720.4	918.5	−21.6%	51.7	57.2	−9.7%

The claim of the previous revision has reversed. Under the old reward the one policy was ahead of all 21 by 14% to 28% of the removal rate. Under the reward of this revision it is ahead by 1.4% to 5.6% over the weights from 0.0 to 0.6, and it is behind at the two heaviest weights, by 9.0% at $w_2 = 0.8$ and by 21.6% at $w_2 = 1.0$. Over the whole range of 0.0 to 0.6 it also spends 2.9% to 6.1% less energy than the per-weight policies, and at 0.8 and 1.0 it spends 8.0% and 9.7% less energy for the throughput it gives up.

The reason is the signal, not the training budget. The weight in the observation is a weaker signal than a dedicated policy where the trade-off is steepest: one conditional policy has to serve 0.0 and 1.0 with the same network, and the sweep of Table 7 shows how far apart those two requests are (8000 RPM against 4111 RPM, 13.1 s against 26.1 s). A policy trained for one weight is not asked to do both. The honest reading is therefore: one conditional policy covers the range and is ahead over most of it, but a policy trained for a single weight is better at the extreme weights, where the trade-off is steepest. The weight stays a runtime query rather than a retrain, but it is not free at the ends of the range.

7.5 Baseline Comparison

The five classical policies are evaluated on the same 50 episodes of the nominal machining condition at the same energy weight $w_2 = 0.4$, so the numbers are comparable. The reward column is the decisive one, because it carries the constraint penalties.

Table 9: Baseline comparison, 50 episodes of the nominal condition at $w_2 = 0.4$

Policy	**Reward**	MRR **(mm^3/min)**	**Energy (kJ)**	**Cycle (s)**	R_a **final (μm)**
Constant high	10.0	**1099.6**	17.8	**4.3**	1.56
NSGA-II static	5.1	732.9	18.8	6.4	1.56
Rule based	3.3	719.8	18.8	6.6	1.56
Receding-horizon	**253.5**	725.5	19.0	6.7	1.25
Constant safe	−1882.9	1.6	385.7	768.0	0.26

- **Three of the four static policies finish outside the roughness safety limit** ($R_{a,\text{safe}} = 0.85 \times 1.6 = 1.36$ μm), at 1.56 μm. A part that finishes out of spec is paid no completion bonus and is charged the finishing-pass penalty, so those three land near zero

(10.0, 5.1 and 3.3) however fast they cut. The fourth, the constant-safe policy, finishes inside the limit at 0.26 μm but needs 768 s and 385.7 kJ per part, which is what its -1882.9 measures.

- **The receding-horizon controller is the strongest baseline** at $+253.5$, and it is the only classical policy that finishes every part inside the limit, at 1.25 μm. It pays for that with throughput and time: 725.5 mm^3/min and 6.7 s against 1099.6 mm^3/min and 4.3 s for the constant-high policy.
- **The learned policy is measured against the same controller in the next two sections**, on 100 randomized scenarios at the same weight, where the controller scores 248.9 and the policy 233.6. On the 50 episodes of this table the figure annotates the mean reward of the learned policy at 248.4, level with the controller's 253.5, and the figure shows the learned policy removing more material per minute.

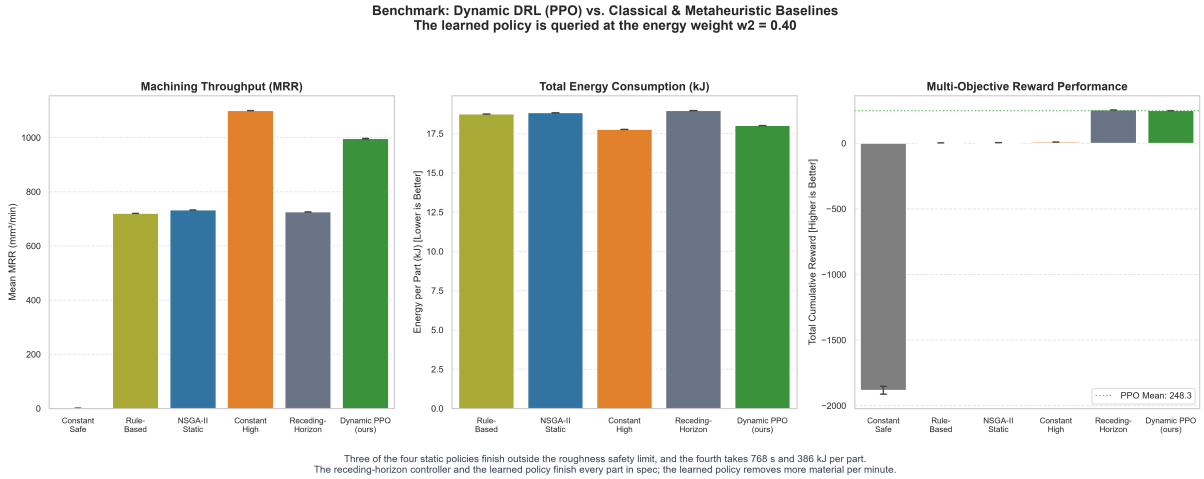


Figure 7: The six policies on the same 50 episodes of the nominal condition at $w_2 = 0.4$: the mean removal rate, the energy per part and the cumulative reward. The four static policies all take a reward near zero or deeply negative, the three fast ones because they finish outside the roughness safety limit and Constant-Safe because it needs 768 s and 385.7 kJ per part, so the comparison that matters is the receding-horizon controller against the learned policy, the only two with a positive reward.

Figure 7 shows the same six policies as three bar charts. The first two panels cannot be compared directly, because Constant-Safe sits at the edge of both scales: it removes 1.6 mm^3/min and needs 385.7 kJ, which is why it appears as a sliver in the throughput panel and is left out of the energy panel. The energy of the other five policies lies between 17.8 kJ and 19.0 kJ, so the energy does not separate them. The third panel does, because the reward carries the constraint penalties: the three fast static policies sit at 10.0, 5.1 and 3.3, the level of the finishing-pass penalty with no completion bonus to cancel it, Constant-Safe sits at -1883, and the receding-horizon controller and the learned policy sit at $+253$ and $+248$. The learned policy removes more material per minute than the controller, and the two are level on the reward.

7.6 Robustness Against the Classical Controller

This is the central result of the work. The two arms are measured on the same 100 randomized scenarios at the same energy weight $w_2 = 0.4$, and the three regimes change what the controller is given, not what the policy is given. The artifact is `results/evaluation/robustness_results.csv` and the script is `experiments/robustness_experiment.py`.

Table 10: Robustness experiment: the same two arms on the same 100 randomized scenarios at $w_2 = 0.4$, in three regimes. Std is the standard deviation of the episode reward over the scenarios.

Regime	**Arm**	**Mean reward**	**Std**	MRR (mm^3/min)	**Energy** (kJ)	**Cycle** (s)	**Tool breakage**	**Parts completed**
Controller has every advantage	Receding-horizon	**248.9**	79.5	1318.2	68.9	17.1	3%	96%
	PPO	233.6	79.7	**1608.5**	66.4	**11.4**	6%	94%
Material model is wrong	Receding-horizon	63.5	117.1	1306.9	71.0	23.3	2%	95%
	PPO	**226.9**	88.3	1604.2	66.3	11.5	6%	94%
Wear measurement is noisy	Receding-horizon	181.3	185.9	1300.4	74.2	31.1	4%	91%
	PPO trained with the noise	**213.6**	87.3	1554.2	66.9	12.0	6%	94%

The fairness decisions come first, because they are what makes the comparison defensible. **Both arms of the table are the general ones:** neither the controller nor the policy was trained for the degraded regime. **The noise is tilted against the policy, not towards it:** the controller is given a noisy reading of the flank wear, which is the measurement it depends on, while the policy receives the noise on the flank wear *and* on the remaining stock. **The noise-trained policy is reported separately and is not the headline:** it exists and it is in the table as a third, clearly labelled arm, but a controller can also be prepared for a noisy instrument by widening its safety margin, and that would be a design choice rather than a training one. Section 7.7 therefore compares the two general arms across the whole range of the noise and reports the noise-trained policy separately.

Four things follow from the table.

1. Inside a perfect digital twin the two methods are comparable, and the measured gap is partly an artifact of the reward. The controller computes the optimum of the same objective at every pass with the exact model, the exact state and the material in its hand, and it scores 248.9 against the policy's 233.6. The gap is inside the noise: the standard error of the mean is about 8 for both arms, so a difference of 15 has a t of about 1.4. And Section 4.5 measures what part of that gap is the reward and not the operating point: the controller cuts the part in 5.85 passes against the policy's 4.00, collects 27% more objective credit for the same material, and at the objective scale of the configuration that is about nine of the fifteen points. A tie is the defensible reading of this row.

2. The learned policy is the better throughput operator. It removes 22% more material per minute (1608 against 1318), finishes a part in a third less time (11.4 against 17.1 s) and spends 3.7% less energy. It pays for that with twice the tool breakage, 6% against 3%, and with 4% of its finished surfaces above the roughness limit against the controller's 2%.

3. The advantage appears where a plant actually is. With a material the controller has no model for, its reward collapses by three quarters, from 248.9 to 63.5, because it does not slow down and its finished surfaces go out of specification. The learned policy holds at 226.9. The same holds for a noisy measurement: with 5% noise on the wear and on the remaining stock the controller's reward falls to 181.3 and its spread more than doubles, to 185.9 against the policy's 87.3. The arm in that row is the policy trained with the same noise, and it keeps 213.6 and cuts the part in 12.0 s against the controller's 31.1 s. The general policy is the arm of the headline comparison, and it scores 226.8 at that same 5% noise.

4. The policy does not need the material to be told. The third arm of the model-error regime, the policy that is told the true material, scores 215.1 with a spread of 94.2, against 226.9 and 88.3 for the policy that believes the nominal material. The material in the observation is a help, not a necessity.

The claim this work supports is therefore narrow and specific: inside a perfect digital twin the learned policy and the model-based controller are comparable, with the learned policy ahead on the physical metrics and behind on the tool breakage, and the gap that the reward does show is partly the artifact of Section 4.5. The advantage of the learned policy appears in the two regimes where the controller's information advantage disappears, a wrong model of the material and a noisy wear measurement. Section 7.7 measures the boundary between the two situations, and the two sweeps of it are one statement: the learned policy is the better operator whenever the setup's model of the machine is wrong by more than about 2%. The comparison against a model-based controller is sensitive to how the reward pays for a rate, and that sensitivity is published with the result rather than left out.

Table 11: Reward sensitivity: the mean reward of both arms on the same 100 randomized scenarios, scored under two objective forms and eight objective scales. The gap is the controller's mean reward minus the policy's, so a positive gap means the controller leads. The shipped form is the sum of the per-pass removal rates; the alternative credits the material of the part against the energy of the part, which removes the pass count from the accounting.

	Shipped form (sum of the per-pass rates)			**Alternative form (material against energy)**		
Objective scale	**Controller**	**Policy**	**Gap**	**Controller**	**Policy**	**Gap**
5	209.7	202.6	+7.1	208.2	201.6	+6.5
20	215.9	207.5	+8.3	209.7	203.4	+6.3
50	228.3	217.3	+10.9	212.9	207.1	+5.8
100	**248.9**	233.6	+15.3	218.2	213.2	+5.0
200	290.1	266.2	+23.9	228.7	225.3	+3.4
400	372.7	331.5	+41.2	249.9	249.6	+0.2
1000	620.2	527.2	+93.0	313.2	322.6	−9.4
4000	1857.9	1506.0	+352.0	630.1	687.4	−57.4

Table 11 is the sensitivity of the ideal case to the reward, and it says three things. **The objective scale is not the lever:** the controller leads at every scale from 5 to 4000 under the shipped form, and the gap only grows with the scale, from 7.1 points at 5 to 352.0 at 4000, so no rescaling of the objective term reverses the sign. **The alternative form was implemented and rejected:** crediting the material of the part against the energy of the part removes the pass count from the accounting, and under it the two arms are much closer (5.0 points at the scale of 100, against 15.3 for the shipped form) and the policy is ahead at the large scales, by 9.4 points at 1000 and 57.4 at 4000. It is not the right form, because the material a part gives up is fixed by its stock, the volume of a part is $\pi D \kappa_{\text{total}} L_{\text{cut}}/1000$ whatever the cutting parameters are, so the throughput weight would be inert and the weight sweep would stop measuring what it claims to measure. **A correct formulation is not obvious:** the removal rate of a part is the ratio of a fixed volume to the part's time, and a rate cannot be accumulated over a part without becoming a volume, so a reward has to either price the cycle time directly or compare the part's average rate to a reference at the end of the part. Both change the meaning of the weights, and the second makes the objective sparse and the credit assignment poor. The reward as shipped therefore carries the artifact, and the artifact is published here.

7.7 The Regime Map: How Wrong the Setup's Model of the Machine Has to Be

The robustness experiment of Section 7.6 compares two situations. The regime map measures the boundary between them, so the answer is a number a plant can act on. The artifact is `results/evaluation/regime_map.csv`, the figure is `results/plots/regime_map.png` and the script is `experiments/regime_map.py`. There are two sweeps, and together they are one state-

ment. The first asks how noisy the wear measurement has to be, the second how miscalibrated the belief about the material has to be.

First sweep: the wear measurement is noisy. The flank wear and the remaining stock are measured with an instrument, not exactly, and the reading carries a Gaussian error of σ, expressed as a fraction of the range of the measured quantity. The controller solves the next pass from that reading, and the policy reads the same noise on its observation.

Table 12: Regime map, first sweep: the mean episode reward of the three arms over the same 100 randomized scenarios at $w_2 = 0.4$, against the measurement noise. The noise is a fraction of the range of the flank wear and of the remaining stock. The last column is the general policy's mean reward minus the controller's, so a positive value means the learned policy is ahead.

Measurement noise	Receding-horizon	PPO	PPO trained for the noise	PPO ahead by
0.0%	**248.9**	233.6	219.6	−15.3
1.0%	242.3	233.4	216.9	−8.9
2.0%	231.5	233.1	214.3	+1.7
3.0%	211.5	230.0	214.1	+18.5
5.0%	181.3	226.8	213.6	+45.5
7.5%	179.1	223.2	207.9	+44.1
10.0%	171.5	222.0	206.5	+50.6

Second sweep: the material model is miscalibrated. The machine runs the material of the scenario, and the setup's belief about that material is off by a systematic fraction, optimistically: it under-rates the wear and the roughness by that fraction, which is the damaging direction, because a setup that thinks its tool is better than it is will push it harder. The same belief goes to both arms, the controller as the model it solves with and the policy as the material in its context.

Table 13: Regime map, second sweep: the mean episode reward of the three arms over the same 100 randomized scenarios at $w_2 = 0.4$, against the miscalibration of the material model. The error is the percentage by which the setup under-rates the wear and the roughness. The last column is the general policy's mean reward minus the controller's, so a positive value means the learned policy is ahead.

Model error	Receding-horizon	PPO	PPO trained for the noise	PPO ahead by
0%	**248.9**	233.6	219.6	−15.3
5%	201.5	232.8	219.6	+31.2
10%	136.0	237.8	219.6	+101.8
20%	35.7	201.5	219.6	+165.8
30%	32.5	116.7	219.6	+84.3
40%	39.3	117.0	219.6	+77.7

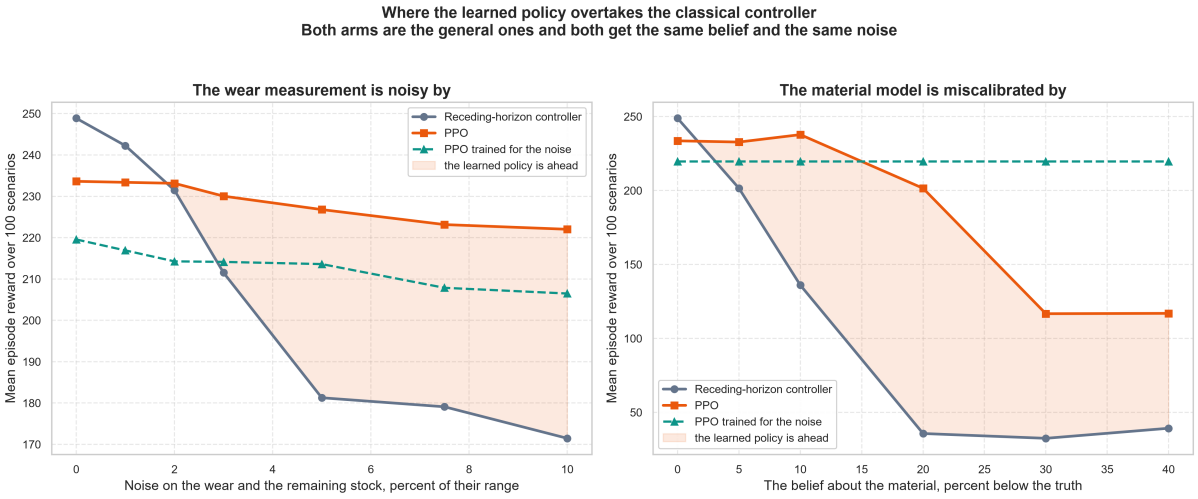


Figure 8: The regime map, two panels, one per sweep. The mean episode reward of the three arms over the same 100 randomized scenarios against the measurement noise on the flank wear and the remaining stock, as a percent of their range (left), and against the miscalibration of the material model (right). The shaded band is where the learned policy is ahead of the controller. The controller's curve falls away in both panels while the policy's curve is nearly flat, so the two cross at about 2% in both.

Three things follow from the first sweep, Table 12 and Figure 8.

1. The crossover is at about 2% of the range of the wear measurement. Below it the classical controller is the better operator, above it the learned policy is. That is the actionable form of the result: a plant that can measure the flank wear to better than 2% of its range has the better controller, and a plant that cannot has the better policy.

2. The two degrade differently. The controller falls from 248.9 at no noise to 171.5 at 10%, a 31% loss, because a noisy wear reading makes its plan violate the limits and break tools. The policy falls from 233.6 to 222.0, a 5% loss, because a perturbed observation perturbs its action slightly. A model-based controller has a cliff, a learned policy has a slope.

3. Training for the noise does not pay. The policy trained with the noise is below the general policy at every noise level, including the 5% it was trained for, where it scores 213.6 against the general policy's 226.8. The robustness of the general policy comes from having been trained across a distribution of scenarios, and training against the noise only makes the policy more conservative.

Of the two degraded regimes of Section 7.6, the noise test is the fairer of the two, because the noise is a property of the instrumentation and not of the method: the material-error regime changes the physics that the controller models, which is a modelling choice, while the noise regime changes only what the instrument reports.

Four more things follow from the second sweep, Table 13.

4. The controller's tolerance of a wrong model is about 2% to 3%, the same order as its tolerance of a noisy measurement. A 5% miscalibration costs it 47 points (248.9 to 201.5) and a 10% error costs it 113 points (to 136.0). It collapses to 35.7 at 20%.

5. The policy is unaffected up to 20%. It scores 201.5 at a 20% error, 13% below its own no-error score of 233.6, and it is marginally better than at zero error at 5% and 10% (232.8 and 237.8 against 233.6, a difference inside the noise). It collapses at 30% (116.7), and the reason is that the belief has then left the distribution the policy was trained on, so its context feature is clipped at the edge of the band.

6. A conservative policy is insensitive to a wrong model. The policy trained for the noise is flat at 219.6 across the whole range, because it keeps the tool away from its limit whatever it believes about the material. It pays for that with the reward, and it is below the

general policy everywhere.

7. So the two sweeps are one statement. The learned policy is the better operator whenever the setup's model of the machine is wrong by more than about 2%, whether that is a wear measurement noisier than 2% of its range or a material model miscalibrated by more than 2%. Below that the classical controller is better. The policy's own limit, around 20% to 25%, is where the belief leaves the distribution it was trained on, and that limit is a design parameter: widen the randomization band of the training and the tolerance widens with it.

The fairness decisions are the same in both sweeps. Both arms of the headline comparison are the general ones, neither is trained for the noise or for the model error. The same belief goes to both arms, so neither is given information the other lacks. The direction of the error is the damaging one, and it is deliberate: the setup under-rates the wear and the roughness, because a setup that thinks its tool is better than it is will push it harder. The policy trained for the noise is reported as a third, clearly labelled arm in both tables, and it is not used for the headline, because a controller can also be prepared for a noisy instrument by widening its safety margin, and that would be a design choice rather than a training one. In the first sweep the noise is tilted against the policy as well: the controller is given a noisy reading of the flank wear, the measurement it depends on, while the policy receives the noise on the flank wear *and* on the remaining stock.

7.8 Batch Against the Part

A tool is a resource that a batch of parts runs out of. The environment `MillingDT/env/milling_batch_env.py` (`MillingDTBatchEnv-v1`) cuts a batch of ten parts with one tool: the tool is replaced when it breaks or reaches its wear limit, the replacement costs 500 reward units against the 200 a finished part is worth, and the observation is the eleven features of the part plus the tool life left and the fraction of the batch still to cut. The artifact is `results/evaluation/batch_results.csv`, measured over 30 batches.

Table 14: Batch experiment: a batch of ten parts cut with one tool, 30 batches. The tool change costs 500 reward units.

Arm	Batch reward	Std	Reward per part	Tool changes	Parts cut	Batch time (s)	Batch energy (kJ)
PPO, batch aware	1817.0	477.8	166.8	**0.63**	10	**113.1**	**610.4**
PPO, per part	1709.9	366.1	169.4	0.83	10	114.7	674.3
Receding-horizon, per part	**1859.9**	328.8	173.7	0.83	10	207.5	692.0

The batch context is worth 24% fewer tool changes and 9.5% less energy at the same completion. The policy that sees the tool life and the batch that is left replaces the tool 0.63 times per batch against 0.83 for the same policy without that context, and it uses 610.4 kJ against 674.3 kJ, and both cut the ten parts. The batch-aware arm also collects more batch reward than the same policy without the context, 1817.0 against 1709.9.

The per-part controller gets the highest batch reward, 1859.9, and it buys it with the machine time. It needs 207.5 s of machine time for the batch against the batch-aware policy's 113.1 s, about 1.8 times the machine time for a batch reward that is 2.4% higher, and its reward per part is 173.7 against the batch-aware policy's 166.8. The reason is structural: the controller solves one part at a time, so it has no notion of a tool that the parts of a batch share, and it cannot price a tool change against the parts that follow it.

7.9 Controller Initialization Experiment

The baseline comparison left one question open: the controller re-solves the next pass with the exact state and the exact physics, so it is a strong teacher, and the learned policy is level with

it on the reward. The experiment asks whether a policy that starts from the controller ends up better than the expert it started from.

The method is a behavior cloning step used as an initialization. The controller's behavior is recorded on 2000 randomized episodes, a supervised actor with the same shape as the PPO actor (two hidden layers of 256 with Tanh) is fitted on those trajectories by regression, and the fitted weights are copied into a fresh PPO policy. The reinforcement learning therefore starts from a feasible behavior instead of from noise. The policy is then trained with the same budget as the production policy (1,000,000 steps) and evaluated on the same 100 randomized scenarios at the same energy weight $w_2 = 0.4$. The clone lives in `MillingDT/agent/controller_init.py`, the experiment in `experiments/controller_experiment.py`, and the trained policy in `results/run_PPO_controller_init_seed42/`.

Table 15: Controller initialization experiment: three arms on the same 100 scenarios at $w_2 = 0.4$. Std is the standard deviation of the episode reward over the scenarios.

Arm	**Mean reward**	**Std**	MRR (mm^3/min)	**Energy** (kJ/part)	**Cycle** (s)	R_a **viol.** finished	**Flank wear violations**	**Tool** breakage	**Parts** completed
Receding-horizon controller	**248.9**	79.5	1318.2	68.9	17.1	2%	1%	3%	96%
PPO from scratch	233.6	79.7	**1608.5**	66.4	**11.4**	4%	0%	6%	94%
PPO initialized from the controller	119.8	178.7	1274.9	81.5	42.6	4%	9%	4%	87%

The initialization is now actively harmful, and the reason is instructive. The policy initialized from the controller scores 119.8 against the from-scratch policy's 233.6, a gap of 113.8, which is more than fourteen times the standard error of the mean over the 100 scenarios. The cloned actor starts the policy at the controller's operating point, which is the conservative one: a low feed and a low spindle speed, so that every pass of the part stays inside the roughness safety limit. With the wear constraint now binding, that conservatism is expensive. The policy then learns from a bad starting point, slows down further to stay away from the wear limit, and ends up cutting a part in 42.6 s against the controller's 17.1 s and the from-scratch policy's 11.4 s, while it spends 81.5 kJ per part against 66.4 kJ. It also exceeds the flank wear limit on 9% of the parts, which the other two arms never do.

The earlier revision found the initialization harmless. It made the policy more consistent and did not change the reward, and the report said so. That was the reward not counting the cost of the slow operating point: a policy that finished every part at any cost looked like a good policy. With the throughput counted, the initialization is what the measurement says it is.

So the method to keep from the controller initialization is the record and clone machinery, not the initialization itself. The recording, the regression and the weight copy are a clean way to put a feasible behaviour into a policy, and they stay in the code, because they are what makes the arms of Table 15 comparable. What the experiment now shows is that starting the reinforcement learning at the controller's operating point is the wrong way to use them.

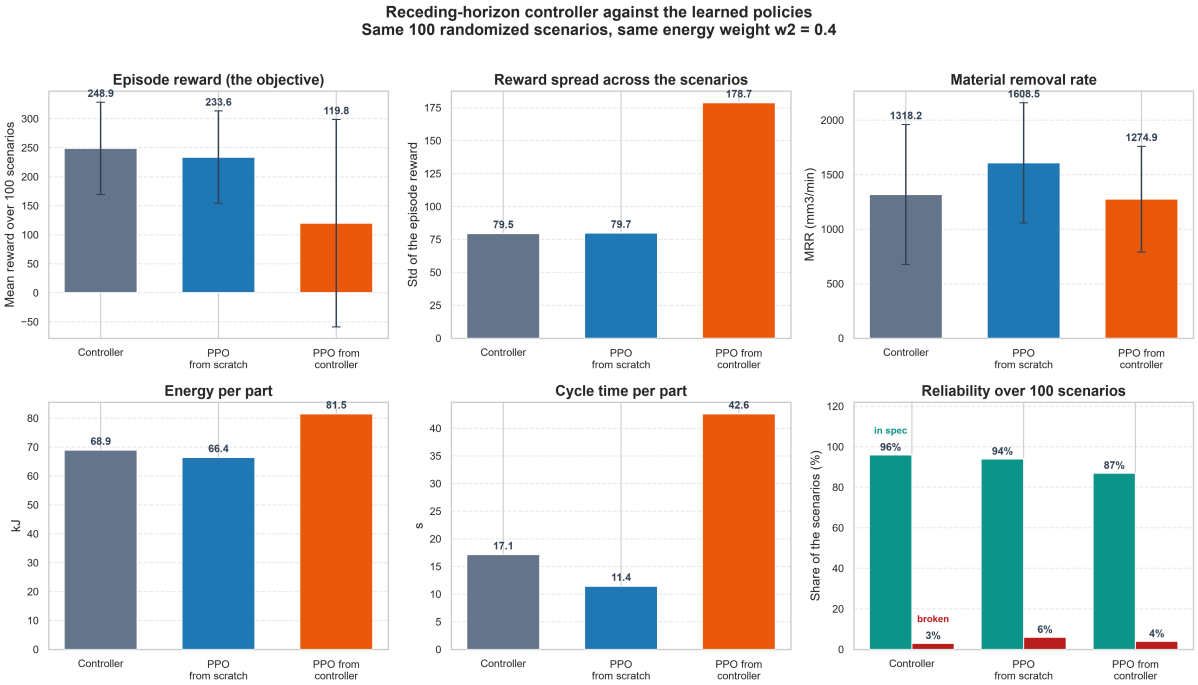


Figure 9: The three arms of the controller initialization experiment on the same 100 scenarios at $w_2 = 0.4$: the mean episode reward with its spread over the scenarios, the spread on its own, the removal rate, the energy per part, the cycle time and the reliability. The initialized arm sits **114 points** below the from-scratch arm on the reward, and its spread of **178.7** is more than twice the spread of the other two. It is the only arm that cuts a part in more than 17 s and the only one that exceeds the flank wear limit.

Figure 9 puts the three arms side by side over the six measures. The top-left panel is the one to read with care: the three bars are 248.9, 233.6 and 119.8, and while the error bars of the first two are far wider than the gap between them, the gap to the third is far wider than its error bar. The second panel shows the same thing in the spread: the reward standard deviation is 79.5 for the controller, 79.7 for the from-scratch policy and **178.7** for the initialized one, so the initialization does not buy consistency any more, it buys variance. The remaining panels show what it costs: the initialized arm removes 1274.9 mm^3/min against 1608.5, spends 81.5 kJ against 66.4 kJ and takes 42.6 s against 11.4 s. On the reliability panel it completes 87% of the parts against 94% and 96%, and it is the only arm with 9% flank wear violations.

7.10 Transfer Experiment

Target material: harder alloy ($k_1 \times 1.25$, $k_2 \times 1.30$) with faster-wearing tool ($C_{\text{wear}} \times 2.0$). The source policy was trained on the nominal material only and has never seen this material.

Table 16: Transfer experiment on the target material, 30 scenarios

Arm	Mean reward	R_a violation worst pass	R_a violation finished surface	Parts completed
Source policy, zero-shot	168.5	100.0%	6.7%	86.7%
Source policy, fine-tuned (50k steps)	101.9	26.7%	0.0%	86.7%
Trained on the target from scratch (50k steps)	101.9	26.7%	0.0%	86.7%
Production policy, zero-shot	**190.1**	100.0%	10.0%	90.0%

The production policy is the best arm without a single adaptation step. It was trained across a distribution of materials with the material context in its observation, so the harder alloy is inside its training distribution. Zero shot it scores the highest reward of the four arms (+190.1) and completes 90.0% of the parts against 86.7% for the other three. Its worst pass violates the roughness limit in every scenario, and its finished surface in 10.0% of them, which is worse on that count than the nominal-trained source policy, and it is still ahead on the reward.

Adaptation repairs the roughness violations and costs more than it gains. Fine tuning the source policy for 50,000 steps on the target, or training on the target from scratch for the same budget, both cut the worst-pass violations from every scenario to a quarter of them and take the finished surface back inside the limit, and both land at exactly the same reward, 101.9, against the source policy's 168.5 zero shot. The reason is the material: the target alloy wears the tool twice as fast, and a policy that slows down for that trades away more throughput than it gains in specification. The lesson is that the cost of a material change is not only the roughness, it is the throughput that the repair gives up.

8 Interactive Digital Twin Workstation

The Streamlit application (`app.py`) provides an industrial-grade interface:

1. **2D Cross-Sectional Stock Depletion:** Real-time visualization of cylindrical billet, pass-by-pass cutout, remaining allowance κ, depth per pass a_p, and tool engagement.
2. **Micro-Topography Generator:** Feed-mark cusp profile with an ISO 1302 centerline average R_a overlay and a 1.6 μm ceiling:

$$y_{\text{cusp}}(x) \approx \frac{x^2}{8r_\epsilon} + \text{stochastic perturbation}$$

3. **ISO 6983 G-Code Synthesizer:** Emits Fanuc/Siemens compliant G-code for the exact PPO parameter sequence.
4. **Multi-Controller Comparison:** PPO against NSGA-II, Rule-Based and Constant-High, with a telemetry overlay.
5. **Cached Policy Loading:** `@st.cache_resource` ensures model loads once per session.

The workstation loads the production policy (`results/run_PPO_seed42`) and exposes the energy weight as a runtime slider, so the whole trade-off is reachable from the browser without a retrain.

Launch: `streamlit run app.py`

9 Software Engineering Practices

9.1 Architecture

- `MillingDT/config/`: Calibrated constants and paths (YAML), the shared physics equations, the scenario distributions (single source of truth)
- `MillingDT/env/`: The Gymnasium environments, two registrations (`MillingDTEnv-v1` for the part, `MillingDTBatchEnv-v1` for the batch)
- `MillingDT/agent/`: The controller initialization, the behavior clone of the receding-horizon controller
- `MillingDT/baselines/`: The receding-horizon controller
- `MillingDT/sim/`: Simulation engine for Streamlit (stock depletion, micro-topography, G-code)
- `MillingDT/utils/`: Seeding, reproducibility
- `experiments/`: The entry points: training, evaluation, ablation, baselines, transfer, controller, robustness, regime map, batch and reward sensitivity experiments, pipeline orchestration
- `viz/`: Plotting utilities, training curves, MC distributions, baseline comparison, controller comparison
- `results/`: One directory per run, holding the policy, the log, the metrics and the figures of that run

9.2 Behavior Cloning as an Initialization

The receding-horizon controller is a competent expert, so it can supply the starting point of the learner instead of leaving the learner to start from noise. The module `MillingDT/agent/controller_init.py` records what the controller does, fits a supervised actor of the same shape as the PPO actor on those trajectories by regression, and copies the fitted weights into a fresh PPO policy. The initialization stays a separate concern: the clone knows nothing about the training loop, the copy is a weight assignment between two tensors of the same shape, and the rest of the run is an ordinary PPO training, selected with the `--init-from-controller` flag. The same separation is what makes the arm comparable: the three arms of Table 15 share the environment, the scenarios, the budget and the evaluation code, so the only difference between them is where the weights start. The measurement of Section 7.9 is that the difference matters: the initialization now costs 113.8 reward points, so the machinery is kept and the initialization is not.

9.3 Reproducibility

- Global seeding via `MillingDT/utils/seeding.py` (Python, NumPy, PyTorch, CUDA deterministic)
- `MillingDT/config/scenarios.py` shared by training and Monte Carlo – no drift
- `experiments/summarize_results.py` rebuilds `RESULTS.md` from committed CSVs
- The comparison tables are committed as CSVs under `results/evaluation/`: the baselines, the ablation summary, the weight sweep against the ablation, the transfer, the controller initialization, the robustness, the regime map, the reward sensitivity and the batch tables

- All random draws (scenarios, weights, seeds) are deterministic given the seed

9.4 Testing

45 tests in `tests/` covering:

- Environment: action clipping, truncation, soft norm, the 11-dimensional observation, physics constants, overrides, info dict, energy norm, Gym checker, randomization, pinned values, scenario scales, the context (weight and material), the belief of the material, the measurement noise, random weight, cutting energy speed dependence, part energy interior minimum, shared physics, shared scenario tables
- Reward: golden-value test (independent recomputation), finishing-pass penalty
- Batch environment: the 13-dimensional batch observation, the tool life and the batch left, every part of the batch, the tool change, the wear carried over, a broken tool, the wear limit, the batch reward, the seeded reproducibility, the pinned weight, Gym checker
- Integration: training→evaluation roundtrip, one registration, the context in the observation, one policy answering a pinned weight, baseline artifact
- Progress callback: heartbeat delivery, failure reporting, interval respect

9.5 Code Quality

- Centralized constants (`MillingDT/config/physics_constants.yaml`), one calibrated value per line with its unit in the file
- Centralized paths (`MillingDT/config/paths.yaml`) – no hardcoded strings
- Environment ID exported from `MillingDT/env/__init__.py`
- No unused imports (AST-verified)
- Conventional commits, clean git history

10 Discussion

10.1 What Worked

1. The speed-dependent cutting energy model ($k_c \propto v_c^m$) is essential. In an earlier revision of the reward, without it the energy/throughput trade-off vanished and the 21 ablation policies came out identical.
2. The scale of the reward matters as much as the weights. With the objective term worth about 1.2 points against a completion bonus of 200, the reward did not count the throughput and the policy learned to finish every part at any cost. At about fifty points per part the trade-off is inside the reward, and the operating point of the production policy changed completely: 1608 mm^3/min against 1205 before.
3. Domain randomization + shared scenario tables: the policy generalizes to unseen scenarios (94% part completion, 4% finished-surface violations).
4. Putting the energy weight in the observation reduces the ablation cost from 21 training runs to one. The one policy is ahead of the per-weight policies over the weights from 0.0 to 0.6 and behind at 0.8 and 1.0, so the weight stays a runtime query, but it is not free at the ends of the range.

5. Training across a distribution of materials removes the need for an adaptation step: on a harder alloy the production policy is the best of the four transfer arms zero shot, while a policy trained on the nominal material has to be repaired.

6. A controller that maximizes the expected episode reward is a far stronger baseline than one that maximizes the objective term per pass: on the same scenarios it went from 184.3 to 248.9, and building it exposed that the earlier baseline comparison was hollow.

7. The robustness experiment is the central result of the work: inside a perfect digital twin the two methods are comparable, the learned policy is the better throughput operator, and its advantage appears exactly where the controller's information advantage disappears, a wrong model of the material and a noisy wear measurement.

8. The regime map turns that comparison into a number a plant can act on, and its two sweeps are one statement: the learned policy is the better operator whenever the setup's model of the machine is wrong by more than about **2%**, whether that is a wear measurement noisier than 2% of its range or a material model miscalibrated by more than 2%. The controller's tolerance of both is 2% to 3%. The two degrade differently (a 31% loss for the controller against 5% for the policy over the noise sweep), training against the noise does not pay, and the policy's own limit of around 20% to 25% of model error is set by the randomization band of its training.

9. Measuring the reward instead of assuming it found an artifact that a fair-looking comparison would have hidden. The reward pays the removal rate of every pass, so the credit a plan collects grows with the number of passes it takes, and the controller collects 27% more objective credit for the same material. The ideal case is a tie once that is accounted for, and the sensitivity is published with the result.

10. The batch context is worth something: 24% fewer tool changes and 9.5% less energy at the same completion.

11. The test suite catches physics regressions (energy interior minimum, cutting energy speed dependence) and reward regressions (golden-value test).

10.2 Limitations

1. Inside a perfect digital twin the learned policy and the model-based controller are comparable, and the work does not claim a reinforcement learning advantage there. The controller scores 248.9 against the policy's 233.6 (Table 10) and the gap of 15 is inside the noise, and about nine of those points are the reward's artifact: the controller cuts the part in more passes and is paid for each of them (Section 4.5). The learned policy is the better throughput operator (22% more material per minute, a third less time, 3.7% less energy) and it pays for that with twice the tool breakage and with more finished surfaces outside the roughness limit. The reading of the ideal case is a tie, not a win.

2. The crossover at about 2% of the range of the wear measurement is a property of this controller, this reward and this noise model. It is measured against a controller that uses the reading directly and takes no safety margin for it, and the arm where the controller widens its margin for a noisy instrument was deliberately excluded from the headline comparison, because that is a design choice rather than a training one. The noise itself is Gaussian and applies to the two measured quantities only, so the boundary is a property of this digital twin and not of a plant until the instrument is measured. The second sweep has the same caveat: the belief is miscalibrated by one systematic fraction applied to the whole material, which is a modelling choice and not a measurement of a real setup.

3. **The policy has a limit of its own, and it is plain in the data.** The learned policy is unaffected by a material model miscalibrated up to 20% (201.5 against its no-error 233.6) and collapses at 30% (116.7), because the belief has then left the distribution the policy was trained on and its context feature is clipped at the edge of the band. The learned policy is therefore not a general answer to a wrong model. It is the better operator up to a miscalibration of about 20% to 25%, and that limit is a property of the training distribution, not of the method.

4. Initializing the policy from the controller is now harmful rather than harmless: 119.8 against the from-scratch policy's 233.6 (Table 15). The clone starts the policy at a conservative operating point, and with the wear constraint binding that conservatism is expensive.

5. The reward pays a rate per pass, and the credit a part collects therefore grows with the number of passes it takes. This is a property of the reward, not of either method, and it biases the comparison against the policy. The scale of the objective term is not the lever (the sign of the gap never flips from scale 5 to scale 4000), but the form of the accounting is: the part level alternative was implemented and rejected because the material a part gives up is fixed by its stock, which would make the throughput weight inert.

6. The wear model is anchored on the tool life of one paper and the exponents are the exponents of the wear rate, not the Taylor tool life exponents, so the tool life varies only by a factor of a few across the machine envelope and a faster cut barely shortens it.

7. The energy weight ablation uses 150,000 steps per weight against 1,000,000 for the production policy, so the ablation frontier is a lower bound on what a pinned policy can reach and may shift with more training.

8. The reward charges the energy of a pass, while the reported metric is the energy of a part. A part cut in many shallow passes draws less power per pass but spends more time on the idle machine, so the two can move in opposite directions. Both are reported.

9. Stochastic tool breakage (6% rate) introduces variance; the reward penalty could be shaped more carefully.

10. The micro-topography model is a simplified feed-mark cusp; real chatter and vibration are not modeled.

11. The energy model is the modelling choice of this work, not a result taken from the reference papers, and it is calibrated against a single reference condition.

12. The frontier is steep at the heavy weights: from $w_2 = 0.5$ to $w_2 = 1.0$ the removal rate falls 41% for 17% less energy, and the cycle time rises from 12.7 s to 20.1 s. A decision-maker who needs a large energy reduction pays for it in throughput and in time.

10.3 Future Work

1. **The sensitivity of the comparison to how the reward pays for a rate:** a reward that prices the cycle time directly, or that compares the part's average rate to a reference at the end of the part, would remove the pass count from the accounting. Both change the meaning of the weights, and the second makes the objective sparse and the credit assignment poor, so neither was adopted. The sensitivity of the published result to this choice has to be measured and published alongside it, as it is here, and the reward as shipped carries the artifact until a better form is found.

2. **Against a restricted controller:** the controller re-solves every pass with the exact state and the exact physics, so it is an oracle that a learned policy cannot match from its observation alone. The robustness experiment answers the model-error and the measurement-noise part of that question; what is left is the information gap inside a perfect twin.

3. **The boundary under a prepared controller:** the regime map deliberately excludes the arm where the controller widens its safety margin for a noisy instrument, because that is a design choice rather than a training one. How far that moves the crossover is the next measurement, together with the crossover measured on the noise of a real instrument rather than on a Gaussian model of it.

4. **Widening the randomization band of the training:** the policy's own limit, around 20% to 25% of model error, is set by the distribution it was trained on, so the tolerance is a design parameter rather than a property of the method. Training across a wider band of material miscalibration is the measurement that would show the tolerance widening with the band, and it is the practical lever a plant has over that limit.

5. **Sim-to-real:** Calibrate physics coefficients on real machine data; add sensor noise and delay to the environment.

6. **Multi-objective RL:** Replace scalarization with Pareto-conditioned networks or hypervolume maximization.

7. **Runtime energy budget:** Drive the production policy with an optimizer that sets w_2 from an energy budget. The weight is a query, not a retrain.

8. **Batch-level planning:** let the policy plan a whole batch instead of a part, with the tool life and the tool change in the objective, so that the context of Section 7.8 is optimized rather than merely read.

9. **Full 3D Digital Twin:** Extend the 2D cross-section to 3D toolpath visualization with collision checking.

11 Conclusion

This project demonstrates a complete pipeline for DRL-based CNC milling optimization:

1. A physics-grounded Digital Twin with speed-dependent cutting energy, tool wear, surface roughness, and stochastic breakage. The wear coefficient is anchored on a tool life rather than on a wear equation, because none of the three reference papers carries one.

2. Two Gymnasium environments, one for a part and one for a batch of ten parts cut with one tool, with domain randomization, an always 11-dimensional conditioned observation and an energy-aware reward whose objective terms are scaled to a quarter of the value of a part.

3. PPO policies trained with proper experimental rigor: Monte Carlo evaluation with confidence intervals, an ablation study mapping the empirical Pareto frontier, and one production policy covering the frontier in a single run.

4. Strong classical baselines including a receding-horizon controller that re-optimizes at every pass and maximizes the expected episode reward over the same physics.

5. A robustness experiment, the central result of the work: the learned policy and the controller on the same 100 scenarios in three regimes. Inside a perfect digital twin the two

are comparable (248.9 against 233.6, a gap that is inside the noise and about nine points of which are the reward's artifact, since the controller cuts the part in 5.85 passes against the policy's 4.00 and is paid for each pass). With a wrong model of the material the controller's reward collapses to 63.5 while the policy holds at 226.9; with a noisy wear measurement the controller falls to 181.3 and the policy keeps 213.6. The learned policy is the better throughput operator in every regime: 22% more material per minute, a third less time and 3.7% less energy, at twice the tool breakage.

6. A regime map with two sweeps, the measurement noise and the material model error, which are one statement: the learned policy is the better operator whenever the setup's model of the machine is wrong by more than about **2%**. The noise sweep crosses over at about 2% of the range of the flank wear measurement, where the controller loses 31% of its reward and the policy 5%; the model error sweep crosses over at 2% to 3%, where a 10% error costs the controller 113 points and the policy still scores 237.8 against 233.6 at no error. The policy's own limit, around 20% to 25% of model error, is set by the randomization band of its training, and the policy trained against the noise is below the general policy at every noise level, including the 5% it was trained for.

7. A reward sensitivity experiment that measures the artifact instead of arguing about it: the reward pays the removal rate of every pass, so the credit a plan collects grows with the number of passes it takes, and the controller collects 27% more objective credit for the same material. The objective scale is not the lever (the gap grows from 7.1 points at scale 5 to 352.0 at scale 4000 and never changes sign), the part level alternative was implemented and rejected because the material a part gives up is fixed by its stock, and the reward as shipped carries the artifact with the sensitivity published next to the result.

8. A batch experiment: a batch of ten parts cut with one tool, where the tool change costs 500 reward units. The batch context is worth 24% fewer tool changes and 9.5% less energy at the same completion, and the per-part controller needs 207.5 s of machine time for the batch against the batch-aware policy's 113.1 s.

9. A controller initialization experiment, which records the receding-horizon controller on 2000 episodes, clones its actor into a fresh PPO policy and trains it with the same 1,000,000-step budget. The result reversed: the initialization is now harmful (119.8 against the from-scratch policy's 233.6), because the clone starts the policy at the controller's conservative operating point and the wear constraint makes that conservatism expensive. The record and clone machinery stays, the initialization does not.

10. A transfer experiment on a harder alloy, which finds that a policy trained across a distribution of materials needs no adaptation there, while a policy trained on the nominal material has to be repaired, and that the repair costs more throughput than it gains in specification.

11. An interactive workstation for visualization, G-code generation, and multi-controller comparison.

12. Full reproducibility: every result table is generated from committed artifacts by a single script.

The key scientific finding on the physics side is that **the speed-dependent specific cutting energy creates a genuine energy/throughput trade-off**, and a DRL agent can learn to navigate it pass-by-pass while respecting surface quality and tool life constraints. Across the measured frontier, from $w_2 = 0.5$ to $w_2 = 1.0$, the removal rate falls 41% while the energy of the part falls 17%, and the spindle speed falls with the energy weight, which is the mechanism

the model predicts. One policy with the weight in its observation makes the whole trade-off accessible at runtime: it is ahead of the 21 policies trained per weight over the weights from 0.0 to 0.6, by 1.4% to 5.6% of the removal rate, and behind at 0.8 and 1.0, by 9.0% and 21.6%. The weight in the observation is a weaker signal than a dedicated policy where the trade-off is steepest.

The result on the classical side is narrower and more interesting than an advantage, and it is a boundary rather than a claim. **Inside a perfect digital twin the learned policy and the model-based controller are comparable, and the honest reading is a tie rather than a win for either of them.** The controller computes the optimum of the same objective with strictly more information: it has the exact model, the exact state and the material in its hand. What the learned policy does have is the throughput: it removes 22% more material per minute, finishes a part in a third less time and spends 3.7% less energy, and it pays for that with twice the tool breakage and with more finished surfaces outside the roughness limit. Part of the gap the reward does show is not an operating point at all: the reward pays the removal rate of every pass, so a plan that cuts the same part in more, shallower passes is paid more, and the controller takes 5.85 passes against the policy's 4.00. The advantage of the learned policy appears where the information advantage of the controller disappears, a wrong model of the material and a noisy wear measurement, and in the batch, where the tool life is a resource shared by the parts. The two sweeps of the regime map of Section 7.7 put a number on the first of those situations, and together they are one statement: the learned policy is the better operator whenever the setup's model of the machine is wrong by more than about **2%**, whether that is a wear measurement noisier than 2% of its range or a material model miscalibrated by more than 2%, and the classical controller is the better operator below that, inside a perfect digital twin. The policy's own limit, around 20% to 25% of model error, is where the belief leaves the distribution it was trained on, and that limit is a design parameter rather than a property of the method: widen the randomization band of the training and the tolerance widens with it. The comparison against a model-based controller is sensitive to how the reward pays for a rate, and that sensitivity is published with the result rather than left out, so the comparison is a statement about a design decision as well as about the two operators.

All code, data, and results are available in the repository at `G:/My Drive/IITM/AI in Mfg`.

References

References

[1] X. Mo, M. Xia, S. Mei, X. Hu, "Deep reinforcement learning-based optimization for machining process: Cutting parameters dynamic optimization considering tool condition degradation," *Journal of Manufacturing Processes* 156 (2025) 1006–1031.

[2] P. Wang, Y. Cui, H. Tao, X. Xu, S. Yang, "Machining parameter optimization for a batch milling system using multi-task deep reinforcement learning," *Journal of Manufacturing Systems* 78 (2025) 124–152, doi: 10.1016/j.jmsy.2024.11.013.

[3] P. Wang, Y. You, W. Liu, D. Ding, F. Xu, "Knowledge-guided deep reinforcement learning framework for feature machining step sequencing," *International Journal of Advanced Manufacturing Technology* 140 (2025) 5617–5636, doi: 10.1007/s00170-025-16606-4.

A Reproducibility Commands

All commands run from the project root `G:/My Drive/IITM/AI in Mfg`:

Listing 1: Key reproducibility commands

```
# Run all tests
pytest tests/ -v

# Train the production policy (1M steps, scenario and weight drawn per
    episode)
python -m experiments.train_rl --steps 1000000 --seed 42 --randomize --
    randomize-weight

# Train the production policy with the energy weight pinned
python -m experiments.train_rl --steps 150000 --seed 42 --randomize --
    w2 0.6

# Train the policy that faces a noisy wear measurement
python -m experiments.train_rl --steps 1000000 --seed 42 --randomize --
    randomize-weight --obs-noise 0.05 --run-name PPO_noisystate_seed42

# Train the batch-aware policy (one tool cuts a batch of ten parts)
python -m experiments.train_rl --env-id MillingDTBatchEnv-v1 --steps
    1000000 --seed 42 --randomize --run-name PPO_batch_seed42

# Monte Carlo sweep of the production policy (100 scenarios)
python -m experiments.evaluate --scenarios 100

# Runtime weight sweep of the same policy (100 scenarios)
python -m experiments.evaluate --sweep-weight --scenarios 100

# Compare the saved weight sweep against the per-weight policies
python -m experiments.evaluate --compare-only

# Ablation study (7 weights x 3 seeds, 150k steps each, weight pinned)
python -m experiments.train_rl_ablation --steps 150000 --scenarios 50
    --slots 7

# Classical baselines
python -m experiments.run_baselines

# Transfer experiment (300k source steps, 50k adaptation steps)
python -m experiments.transfer_experiment 300000 50000

# Controller initialization experiment (1M steps, 2000 controller
    episodes)
python -m experiments.controller_experiment 1000000 2000

# Robustness experiment (three regimes against the classical controller
    )
python -m experiments.robustness_experiment

# Regime map (the noise sweep and the model error sweep that find the
    crossover)
python -m experiments.regime_map

# Batch experiment (the batch against the part)
python -m experiments.batch_experiment

# Reward sensitivity (the objective scale and the objective form)
python -m experiments.reward_sensitivity
```

```
# Regenerate all plots
python -m viz.plot_results
python -m viz.plot_eval
python -m viz.plot_baselines_comparison
python -m viz.plot_controller_comparison

# Regenerate RESULTS.md from artifacts
python -m experiments.summarize_results

# Launch interactive workstation
streamlit run app.py

# Full pipeline (sequential, eleven stages)
python -m experiments.run_pipeline --steps 1000000 --ablation_steps
   150000
```

B Project Structure

The project holds two Gymnasium registrations, `MillingDTEnv-v1` for the part and `MillingDTBatchEnv-v1` for the batch, one training script and one evaluation script. The registrations are exported as the constants `MillingDT.env.ENV_ID` and `MillingDT.env.BATCH_ENV_ID`. There is no separate conditioned environment, no separate conditioned training script and no `results/conditioned/` directory. The reusable code lives in the `MillingDT/` package, the entry points in `experiments/`, and every training configuration keeps its own directory under `results/`.

Listing 2: Project directory tree

```
G:\My Drive\IITM\AI in Mfg
|-- MillingDT/                        # The library package
|   |-- __init__.py                    # Exports ENV_ID, ALL_CONSTANTS
|   |-- config/
|   |   |-- __init__.py                # Loads the two YAML files
|   |   |-- physics_constants.yaml     # The calibrated constants and the
    reward scales
|   |   |-- paths.yaml                 # Where the runs are kept
|   |   |-- physics.py                 # Shared physics equations
|   |   `-- scenarios.py               # Randomized scenario sampling
|   |-- env/
|   |   |-- milling_env.py             # Gymnasium environment (11-D
   observation)
|   |   |-- milling_batch_env.py       # Batch environment (13-D
   observation)
|   |   `-- __init__.py                # Exports ENV_ID, BATCH_ENV_ID
|   |-- agent/
|   |   |-- controller_init.py         # Behavior clone of the controller
|   |   `-- __init__.py
|   |-- baselines/
|   |   |-- receding_horizon.py        # Receding-horizon controller
|   |   `-- __init__.py
|   |-- sim/
|   |   |-- machining_sim.py           # Stock depletion, topography, G-
   code
|   |   `-- __init__.py
|   `-- utils/
|       |-- seeding.py                 # Global deterministic seeding
|       `-- __init__.py
```

```
|-- experiments/                        # The entry points
|   |-- train_rl.py                     # The one training script
|   |-- train_rl_ablation.py            # Energy weight ablation, pinned
|   |-- evaluate.py                     # Monte Carlo sweep, weight sweep
|   |-- run_baselines.py                # 5 classical baselines
|   |-- transfer_experiment.py          # Cross-material transfer
|   |-- controller_experiment.py        # Controller initialization, 3
   arms
|   |-- robustness_experiment.py        # Three regimes against the
   controller
|   |-- regime_map.py                  # The two sweeps that find the
   crossover
|   |-- batch_experiment.py             # The batch against the part
|   |-- reward_sensitivity.py           # The objective scale and the
   objective form
|   |-- run_experiments.py              # Parallel training supervisor
|   |-- run_pipeline.py                 # Full pipeline orchestrator
|   |-- summarize_results.py            # Regenerates RESULTS.md
|   `-- __init__.py
|-- viz/
|   |-- plotting_utils.py               # Shared plotting functions
|   |-- plot_eval.py                    # Monte Carlo distributions
|   |-- plot_results.py                 # Training curves
|   |-- plot_baselines_comparison.py # Baseline comparison charts
|   |-- plot_controller_comparison.py # Controller against the learned
   policies
|   `-- __init__.py
|-- tests/
|   |-- test_env.py                     # Environment and reward tests
|   |-- test_batch_env.py               # Batch environment tests
|   |-- test_training.py                # Training tests
|   |-- test_progress_callback.py       # Callback tests
|   `-- test_integration.py             # Integration tests
|-- results/                            # One directory per configuration
|   |-- run_<config>/                   # models/, log/, metrics/, plots/
|   |-- evaluation/                     # The tables that compare runs
|   |-- plots/                          # The cross-run figures
|   `-- RESULTS.md
|-- app.py                              # Streamlit workstation
|-- AUDIT_REPORT.md                    # Code audit with dispositions
|-- PROJECT.md                         # Consolidated documentation
|-- README.md                          # Project overview
`-- report/
    `-- report.tex                     # This file
```

C Hyperparameter Sensitivity

The PPO hyperparameters in Table 4 are the Stable-Baselines3 defaults, with three exceptions: the network is widened to 2×256 units, the entropy coefficient is raised to 0.02 to keep the policy from collapsing to deterministic actions early, and the training runs on the CPU with one thread per process so that several policies can train in parallel on one machine. Measured on this machine, PPO on this MLP runs at 812 steps/s on the CPU against 318 steps/s on the GPU, so the GPU is not used.

The entropy coefficient matters most: with too little of it the policy settles into a deterministic action early and stops exploring the envelope.

D Energy Model Derivation

The specific cutting energy $k_c(v_c)$ models the empirical observation that the energy required to remove a unit volume of material increases with cutting speed due to:

1. Higher shear zone temperatures → increased flow stress
2. Increased strain rates → strain-rate hardening
3. Tool-chip interface friction changes

The power-law form $k_c(v_c) = k_{c0}(v_c/v_{\text{ref}})^m$ with $m \approx 0.6$ is a standard empirical form for the rise of the specific cutting energy with cutting speed, and it is the modelling choice of this work rather than a result taken from either reference paper: the reference papers optimize quality and efficiency, and treat energy only as related work. The reference condition v_{ref} is the cutting speed at the calibrated nominal parameters ($n = 5000$ RPM, $D = 50$ mm → $v_{\text{ref}} \approx 785$ m/min). At this speed, $k_c = k_{c0} = 2.5$ W·min/mm^3.

Total power: $P_{\text{total}} = P_{\text{idle}} + k_c(v_c) \cdot \text{MRR}$. Energy per pass: $E_{\text{pass}} = P_{\text{total}} \cdot t_{\text{cut}}$. Substituting $t_{\text{cut}} = L_{\text{cut}}/(nf_r)$ and $\text{MRR} = v_c a_p f_r$:

$$E_{\text{pass}} = P_{\text{idle}} \frac{L_{\text{cut}}}{nf_r} + k_{c0} \left(\frac{v_c}{v_{\text{ref}}} \right)^m v_c a_p f_r \frac{L_{\text{cut}}}{nf_r} \tag{14}$$

$$= \underbrace{P_{\text{idle}} \frac{L_{\text{cut}}}{nf_r}}_{\text{Idle energy}} + \underbrace{k_{c0} \frac{L_{\text{cut}} a_p}{nf_r} \left(\frac{\pi D n}{1000 v_{\text{ref}}} \right)^m \frac{\pi D n}{1000}}_{\text{Cutting energy}} \tag{15}$$

The idle term $\propto 1/(nf_r)$ decreases with speed; the cutting term $\propto n^m$ increases with speed. Their sum has an interior minimum, which is the operating point selected by w_2.